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SUPERSEDING
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Amendment 2
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MILITARY SPECIFICATION

CARRIERS, ARMORED, FULL TRACKED, M901 TOW VEHICLE (LESS TOW WEAPON) AND M981 FISTV; PROCESSING FOR STORAGE AND SHIPMENT OF

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the processing of the M901 and M901A1 Combat Vehicle, Anti-Tank, Improved TOW Vehicle (less TOW Weapon) and the M981 Fire Support Team Vehicle (FISTV) for storage and shipment.

1.2 Classification. Processing shall be of the following levels, as specified (see 6.2):

Level A - Maximum military protection

Level A is the processing required for the protection of vehicle during shipment, handling, and storage exceeding 90 days from date of actual processing. This level does not provide for driveaway capability. It does provide for domestic or overseas shipment, including open deck loading.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GSS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

NO DELIVERABLE DATA REQUIRED AREA PACK

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Level B - Minimum military protection

Level B is the limited processing required for the protection of vehicles during shipment, handling, and storage not to exceed 90 days from date of actual processing. This level provides for driveaway capability, when specified, and domestic or overseas shipment (excluding open deck loading).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
FEDERAL

A-A-883	- Tape, Pressure Sensitive Adhesive, Masking.
O-I-490	- Inhibitor, Corrosion, Liquid Cooling System.
O-S-576	- Sodium Bicarbonate, Technical.
O-S-801	- Sulfuric Acid, Electrolyte, for Storage Batteries.
P-D-220	- Detergent, General Purpose.
GG-T-250	- Tester, Antifreeze Solutions.
NN-P-530	- Plywood, Flat Panel.
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet.
QQ-S-781	- Strapping, Steel and Seals.
QQ-W-470	- Wire, Steel, Carbon, Spring, Music.
RR-W-360	- Wire, Fabric, Industrial.
TT-E-529	- Enamel, Alkyd, Semigloss.
TT-I-1795	- Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces).
TT-V-121	- Varnish, Spar, Water Resisting.
UU-P-268	- Paper, Kraft, Wrapping.
UU-T-81	- Tags, Shipping and Stock.
VV-F-800	- Fuel, Oil, Diesel.
VV-L-800	- Lubricating Oil, General Purpose, Preservative, (Water Displacing, Low Temperature).
ZZ-T-416	- Tire, Pneumatic: Retread and Repair Materials.
LLL-B-810	- Building Board, (Hard Board) Hard Pressed, Vegetable Fiber.
MMM-A-178	- Adhesive, Paper Label, Water-resistant.
NNN-P-40	- Paper, Lens.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Box, Wood, Nailed and Lock Corner.
PPP-B-636	- Box, Shipping, Fiberboard.

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PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-C-1752	- Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible.
PPP-F-320	- Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-S-760	- Strapping, Nonmetallic (and Connectors).
PPP-T-60	- Tape, Packaging, Waterproof.
PPP-T-97	- Tape, Packaging/Industrial, Filament Reinforced.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-117	- Bags, Sleeves and Tubing-Interior Packaging.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-C-450	- Coating Compound, Bituminous Solvent Type, Black (for Ammunition).
MIL-R-3065	- Rubber, Fabricated Products.
MIL-C-5501	- Caps and Plugs, Protective, Dust and Moisture Seal.
MIL-C-5501/7	- Caps and Plugs, Protective, Dust and Moisture Seal (Cap-Plug, General Purpose).
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-B-11188	- Batteries, Storage: Lead-Acid.
MIL-A-11755	- Antifreeze, Artic-Type.
MIL-B-12841	- Basic Issue Items for Military Vehicles, Carriages and Equipment, Preparation for Shipment and Storage of.
MIL-P-13983	- Paint, Temporary, Lusterless, Gasoline Removable.
MIL-P-14232	- Parts, Equipment and Tools for Army Materiel, Packaging of.
MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-D-16791	- Detergent, General Purpose (Liquid, Nonionic).
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-T-22085	- Tapes, Adhesive, Preservation and Sealing.
MIL-B-22191	- Barrier Materials, Transparent, Flexible, Heat-Sealable.
MIL-P-46002	- Preservative Oil, Contact and Volatile Corrosion-Inhibited.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-T-50036	- Talc, Technical, T1 and T3.
MIL-H-52079	- Hose, Preformed: Semi-Flexible, Reinforced.
MIL-D-81298	- Dye, Liquid, for the Detection of Leaks in Aircraft Fuel Systems.

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STANDARDS
MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
AN960	- Washer, Flat.
MS35842	- Clamp, Hose: Low Pressure Type "F".
MS51971	- Nut, Plain, Hexagon-Steel, Corrosion Resisting, 300 Series, Passivated, UNC-2B.
MS90725	- Screw, Cap, Hexagon Head, (Finished Hexagon Bolt) Steel, Grade 5, Cadmium Plated, UNC-2A

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

12269239	- Closure Kit.

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Carrier Safety Regulations

(Application for copies should be addressed to the Department of Tranpsortation, Federal Highway Administration, Washington, DC 20420.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASSOCIATION OF AMERICAN RAILROADS PUBLICATIONS

Section No. 1	- General Rules Governing Loading of Commodities on Open Top Cars.
Section No. 6	- Rules Governing the Loading of Department of Defense Material on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren, Chicago, Illinois 60605.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), one of the first 10 production processed vehicles shall be subjected to the first article inspection specified in 4.2. Approval of this vehicle shall not relieve the contractor of his obligation to process all vehicles in accordance with this specification. Unless otherwise specified by the acquisition activity, any change to materials or design after approval shall require additional vehicles be inspected as specified in 4.2. Government representative shall select the vehicle to be inspected.

3.2 Materials. Materials shall be as specified herein and on the standards, specifications, and drawings referenced herein (see 4.5.1).

3.3 Government furnished equipment. Unless previously accomplished, Government furnished equipment (GFE) (other than installed) shall be packaged, packed and marked in accordance with the individual document for the specific item. GFE shall be stowed with basic issue items (BII).

3.4 Preservatives and atomized spray equipment. When atomized spraying of preservative oil is specified, all compressed air supply lines shall be equipped with moisture separators.

3.5 Processing. Processing the vehicles for level A and B shall be as specified below. Level B processing is the same as level A except for differences as specified.

3.5.1 Preparation prior to processing. Except as otherwise specified herein, and to the maximum extent consistent with production efficiency, economy, and safe storage and shipment, the vehicle shall be prepared for storage and shipment in a completely assembled condition after test runs, and completion and approval of all necessary repairs. Specified equipment shall be installed and all adjustments made so that the vehicle may be operated, shipped, and placed into service with a minimum of delay.

3.5.1.1 Processing records. Records of vehicle processing shall be maintained and shall be readily available for review by Government representatives.

3.5.2 Disassembly. Projecting parts whose removal will accomplish desired cube reduction, and parts susceptible to damage and pilferage, shall be removed from the vehicle. Removed bolts, nuts, screws, pins, and washers shall be placed in one of the mating parts and secured. Removed parts shall be packaged, identified, and stowed securely within the vehicle.

3.5.2.1 Matchmarking. Parts removed from the vehicle shall be matchmarked when necessary to facilitate reassembly. Matchmarking information shall be put on cloth shipping tags conforming to type A of UU-T-81 or on metal tags using waterproof ink or paint, and attached to mating parts. The marked cloth shipping tags shall be waterproofed with varnish conforming to TT-V-121 or adhesive conforming to MMM-A-178.

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3.5.3 Record forms. Two copies of DD Form 1397 shall be provided. Information on forms shall include preservation accomplished and depreservation instructions. The Equipment Log Book Binder and one copy of DD Form 1397 (see 6.4) shall be placed in a bag conforming to type I, class B, style 2, 6 mil of MIL-B-117; the bag shall be closed by heat sealing and securely attached in driver compartment of the vehicle. The other copy of DD Form 1397 shall be water-proofed with adhesive conforming to MMM-A-178 or placed in a bag confoming to type I, class B, style 2 of MIL-B-117. The bag shall be closed by heat sealing and securely attached in a conspicuous location on the exterior of the vehicle.

3.5.4 Cleaning and drying (see 4.5.2.1).

3.5.4.1 Vehicle interior. Interior surfaces of the vehicle shall be cleaned with a solution of detergent conforming to P-D-220 or type I of MIL-D-16791 and water. The solution temperature shall not exceed 210 degrees Fahrenheit (°F) [100 degrees Celsius (°C)] and the pressure shall not exceed 5 pounds per square inch (psi) [35 kiloPascals (kPa)] measured 4 inches [10 centimeters (cm)] from the nozzle. After cleaning, the surfaces shall be rinsed with clean water and dried. Care shall be taken during cleaning and rinsing operations to assure that no solution or water enters the instruments, connections, or other components susceptible to water damage. The solution or water shall not accumulate and remain in cavities that cannot be drained. Vehicles with decals, markers, and straps and floor plates installed shall only be hand cleaned with a solution of P-D-220 or type I of MIL-D-16791 and water to prevent damage to these components. The cleaned surfaces shall be hand rinsed and dried.

3.5.4.1.1 Battery supports and retainers. The battery supports and retainers shall be cleaned with a solution of one-half pound (lb) [227 grams (gm)] of sodium bicarbonate conforming to O-S-576 per gallon [4 liters] of water. After cleaning, the surfaces shall be flushed with clean water and thoroughly dried. Dried surfaces shall then be preserved in accordance with 3.5.5.2.

3.5.4.1.2 Backrests and seats. Backrests and seat cushions shall be cleaned with a solution of detergent conforming to P-D-220 or type I of MIL-D-16791, in warm water. After cleaning, cushions shall be wiped with cloth saturated with clean water to remove cleaning solution. Care shall be taken not to saturate the cushions with cleaning solution or water. After rinsing, cushions shall be dried and protected in accordance with 3.5.6.3.

3.5.4.2 Vehicle exterior and weapon/targeting station. The vehicle exterior and weapon/targeting station shall be cleaned with a solution of detergent conforming to P-D-220, or type 1 of MIL-D-16791, in warm water or steam. Cleaning shall remove all foreign matter. After cleaning, surfaces shall be rinsed with clean water or steam and thoroughly dried. Care shall be taken to avoid entry of water or steam into driver's compartment, engine compartment or weapon/targeting station openings. Care shall be taken in the cleaning of the Image Transfer Assembly (ITA). Steam or high pressure water

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spray shall not be sprayed directly on any of the lenses. All exposed lenses shall be wiped clean and dry using lens tissue conforming to NNN-P-40. Where possible, all lens on the ITA shall have caps placed over them with lens tissue folded and placed inside of each cap. All caps shall be snug fitting and shall be secured in place with tape conforming to type III, class 1, of PPP-T-60.

3.5.5 Preservation.

3.5.5.1 Relubrication. If the vehicle has been operated more than 75 miles [120 kilometers (km)] since lubrication or after the vehicle has been cleaned in accordance with 3.5.4.2, the vehicle shall be relubricated using materials conforming to the drawing, specification, or lubrication order applicable to the vehicle. All exposed oil can points shall be coated with lubricant conforming to VV-L-800. These points include such items as levers, locking levers, locking bars, locking pins, pintle pins, hinge pins, hinges, strikers, wing nuts, door locks, hand-operated locking knobs, latches, linkage, threaded ends of yokes, and related clevis pins. Excess lubricant shall be removed after coating.

3.5.5.2 Preservation of battery supports and retainers. The top battery supports and retainers shall be preserved with compound conforming to composition G, type I or II of MIL-C-450.

3.5.5.3 Transmission, transfer assembly, controlled differential and final drives. Transmission shall be filled to operating level with lubricating oil conforming to grade 10W of MIL-L-21260. Transfer assembly, controlled differential, and final drives shall be filled to operating level with lubricating oil conforming to grade 10W, or grade 30, of MIL-L-21260 as applicable. DD Form 1397 shall be annotated with the type and grade of lubricant or preservative oil used (see 3.5.3).

3.5.5.3.1 Level B: Transmission, transfer assembly, controlled differential, and final drives. Transmission, transfer assembly, controlled differential, and final drives shall contain operational lubricant, as specified on applicable drawings, specifications, or lubrication order, filled to operating level. If, however, these units contain lubricating oil conforming to grade 10W, or grade 30, of MIL-L-21260, an additional amount of the same oil shall be added to attain the operating level. Operating lubricants shall not be mixed with MIL-L-21260 except in an emergency. DD Form 1397 shall be annotated to indicate the type and grade of operational lubricant or preservative oil used.

3.5.5.4 Engine crankcase preservation. The crankcase shall be filled to the operating level with lubricating oil conforming to MIL-L-21260 of the seasonal grade specified in the applicable drawing, specification, or lubrication order. DD Form 1397 shall be annotated with the type and grade of lubricant used.

3.5.5.4.1 Level B: Engine crankcase preservation. The engine crankcase shall be filled to the operational level with normal operational lubricant, as specified in the lubrication order. DD Form 1397 shall be annotated to indicate the grade of lubricant used.

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3.5.5.5 Engine preservation. The engine preservation shall be an uninterrupted sequence in accordance with 3.5.5.5.1 and 3.5.5.5.2.

3.5.5.5.1 Fuel system and combustion chamber. Prior to processing, the engine shall be cooled to assure that cylinder head temperature, measured at injector nozzle flange surfaces of all cylinders, is not more than 100°F (38°C). Cooling shall be accomplished by induced air currents, circulation of engine coolant, or by waiting the period of time required to attain the above specified temperature. When the ambient temperature exceeds 100°F (38°C), the engine shall be cooled to a temperature equivalent to the ambient. After the engine has been cooled, the fuel supply system from the fuel tank shall be shut off. A portable container with two compartments shall be positioned to provide gravity feed to the engine. One compartment shall be filled with preservative oil conforming to grade 1 of MIL-P-46002 to which has been added an oil soluble red dye conforming to MIL-D-81298, in a concentration sufficient to impart a marked coloring to the oil. The second compartment shall be filled with diesel fuel conforming to VV-F-800. Disconnect the fuel line between the primary fuel filter and fuel pump at the primary fuel filter outlet. Drain the diesel fuel from the secondary fuel filter. Remove the filter can and remove the filter element. Fill the filter can with preservative oil conforming to grade 1, MIL-P-46002, and reinstall it. Connect the fuel line from the auxiliary fuel container to the fuel pump supply line. Disconnect the vehicle fuel return line at the quick disconnect coupling. Connect a transparent plastic fuel line to the engine end of the disconnected fuel return line, insert the other end of the plastic line into a recovery container to collect the returned oil. Disconnect the air cleaner hose from the air inlet elbow. An air restrictor boot fabricated in accordance with figure 1 shall be installed on the air inlet elbow and secured with a clamp to assure that air to the engine is completely cut off. The fuel valve on the auxiliary fuel tank shall be turned to the preservative oil position. Place the engine fuel control in RUN position. Crank the engine with the starter (NOTE: The engine may fire for approximately five seconds) for not less than 30 seconds and not more than 45 seconds. The fuel valve on the auxiliary fuel container shall be turned to the OFF position. Remove the filter can from the secondary fuel filter. Drain the preservative oil from the can and wipe clean with a lint free cloth. Install the removed element into the can and, fill the can with diesel fuel VV-F-800. Reinstall the filter can into the original position.

CAUTION: Special precautions shall be taken to assure that the cranking time limits specified shall not be exceeded or the engine starter or starter solenoid may be damaged. Excessive cranking could also cause hydrostatic lock.

3.5.5.5.2 Engine purging. Purge the fuel lines of all preservative (grade 1, MIL-P-46002) by turning the auxiliary fuel tank valve to the diesel fuel position and close the engine fuel control shut off. Crank the engine with the starter to remove all traces of red colored preservative oil from the plastic return line. (CAUTION: DO NOT OPERATE THE STARTER FOR MORE THAN 45 SECONDS.) If the preservative oil has not been purged, rest the starter

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for three minutes and repeat purge operation. Turn the auxiliary fuel tank valve to the OFF position. Remove the fuel line and reconnect it to the outlet side of primary fuel filter. Turn on the vehicle fuel supply system and remove the air restrictor boot. Disconnect the plastic fuel return line and reconnect the vehicle fuel return line at the quick disconnect coupling.

3.5.5.5.3 Preservation through air intake and exhaust system. After preservation specified in 3.5.5.5.1 and 3.5.5.5.2, one ounce (oz) [30 milliliters (ml)] of preservative oil conforming to grade 1 of MIL-P-46002 shall be atomized into the exhaust opening. The opening shall then be sealed with tape conforming to type IV, class 1 of PPP-T-60 or type II of MIL-T-22085. The hose shall be disconnected at the air intake, and one ounce (30 ml) of preservative oil, conforming to grade 1 of MIL-P-46002, shall be atomized into the air intake opening. The air intake opening and the engine crankcase breathers shall be sealed with plastic plugs conforming to MIL-C-5501, or with tape conforming to type IV, class 1 of PPP-T-60 or type II of MIL-T-22085. A red warning tag, bearing the information "ENGINE PRESERVED WITH VCI - DO NOT CRANK" and "BEFORE CRANKING, REMOVE CAPS, TAPE, OR PLUGS, AND REINSTALL THE HOSE", shall be placed in a conspicuous location within driver's compartment. DD Form 1397 shall be annotated to show the engine is preserved with VCI and oil and openings are sealed.

3.5.5.5.4 Preservation through oil level gage rod opening. After the preservation specified in 3.5.5.5.1 and 3.5.5.5.2, the oil level gage rod shall be removed, and six ounces (177 ml) of preservative oil conforming to grade 1 of MIL-P-46002 shall be atomized into the crankcase through the oil level gage rod opening. An extension of sufficient length to permit the spray nozzle to be within the crankcase shall be used. The spray nozzle shall not be submerged in the crankcase oil. After spraying, the oil level gage rod shall be reinstalled. All openings to the engine interior, including the oil level gage rod opening and the oil filler cap shall be sealed with tape conforming to type IV, class 1 of PPP-T-60 or type II of MIL-T-22085.

3.5.5.5.5 Preservation through flywheel housing. Two ounces (60 ml) of preservative oil conforming to grade 1 of MIL-P-46002 shall be atomized into the flywheel housing.

3.5.5.5.5.1 Level B: Engine preservation. The engine shall require no preservation for level B shipment and storage.

3.5.5.6 Air cleaner. Air cleaner openings shall be sealed with tape conforming to type IV, class 1 of PPP-T-60 or type II of MIL-T-22085.

3.5.5.7 Personnel and engine compartment heaters and lines. The personnel and engine heater fuel supply shut off valve, located at the inlet side of the fuel filters, shall be turned to the OFF position. The main fuel line to the heaters shall be disconnected at a point closest to the shut off valve. Fuel shall be allowed to drain from the fuel lines. Seal the ends of the disconnected fuel lines and shut off valves with plastic plugs/caps conforming to MIL-C-5501 and tape conforming to type IV, class 1, of PPP-T-60, or type II of MIL-T-22085. External exhaust and intake stacks shall have openings sealed with tape conforming to type IV, class 1 of PPP-T-60, or type II of MIL-T-22085. Plastic plug/cap conforming to

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MIL-C-5501 may be used. Four warning tags shall be prepared, each bearing the information "HEATER FUEL LINES DISCONNECTED AND SEALED. PRIOR TO PLACING PERSONNEL OR ENGINE HEATERS IN OPERATION, REMOVE PLUGS/CAPS OR TAPE FROM FUEL LINES, EXHAUST AND INTAKE STACKS AND SHUT OFF VALVES. OPERATE HEATER FUEL PUMP AND DRAIN A MINIMUM OF ONE QUART OR FUEL THROUGH THE FUEL LINES INTO A PORTABLE CONTAINER. RECONNECT HEATER FUEL LINES". One tag each shall be secured to the personnel and engine heater operating switches, and one each to the personnel and engine heaters.

3.5.5.7.1 Level B: Personnel and engine compartment heaters and lines. The personnel heater and fuel pump shall be in a ready-to-use condition. Heater exhaust and intake openings shall be closed with a plastic plug conforming to MIL-C-5501/7. A warning tag, bearing the information "HEATER EXHAUST AND INTAKE OPENINGS CLOSED, REMOVE PLUGS BEFORE OPERATING", shall be attached to the heater controls.

3.5.5.8 Fuel tank preservation. Fuel tank(s) shall be drained to the maximum extent possible. Fuel tank cap(s) and filler screen(s) shall be removed and coated with lubricating oil conforming to grade 30 of MIL-L-21260. One quart (1 liter) of lubricating oil conforming to grade 10W of MIL-L-21260 shall be added to each five gallons (20 liters), or portion thereof, of residual fuel. The contractor shall develop and maintain a written procedure to ascertain the amount of residual fuel. Tank cap(s) and filler screen(s) shall be reinstalled.

3.5.5.8.1 Fuel tank security. After processing the fuel tank(s) as specified in 3.5.5.8, the armored fuel cap(s) shall be secured with the combat lock.

3.5.5.8.2 Level B: Fuel tank preservation. Unless otherwise specified (see 6.2), the vehicle shall be shipped without draining residual fuel from the fuel tank(s).

3.5.5.9 Cooling system. The cooling system shall be protected by one of the following procedures (see 6.2):

a. For shipment and storage in areas where temperatures drop below minus 40°F (-40°C), the systems shall be protected as specified in 3.5.5.9.3.
b. For shipment and storage within the bounds of 30 degrees north latitude and 20 degrees south latitude, except continental United States, systems shall be protected as specified in 3.5.5.9.2.
c. For all other shipments, cooling systems shall be protected as specified in 3.5.5.9.1.

NOTE: DD Form 1397 (see 3.5.3) shall be completed to indicate coolant used.

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3.5.5.9.1 Water and antifreeze procedure. The cooling system shall be filled to operating level with a clean solution consisting of equal parts by volume of antifreeze (ethylene glycol) conforming to MIL-A-46153 and water. The engine shall be operated until a temperature has been reached that causes the thermostat to open to assure complete mixing and even distribution of the antifreeze solution. A warning tag bearing the information, "COOLING SYSTEM FILLED WITH WATER AND ANTIFREEZE SOLUTION (ETHYLENE GLYCOL) IN EQUAL PARTS BY VOLUME - DO NOT DRAIN", shall be securely attached to radiator filler neck.

3.5.5.9.2 Water and corrosion inhibitor procedure. The cooling system shall be filled with clear water up to the radiator upper tank. A corrosion inhibitor conforming to O-I-490 shall be added in the proportion of 5 ounces (150 ml) of the inhibitor for each 10 quarts (10 liters) of water. The inhibitor shall be dissolved in 2 quarts (2 liters) of warm water and poured into the radiator while the engine is idling. More water shall be added if necessary, to fill the radiator to operating level. A warning tag bearing the information, "COOLING SYSTEM DOES NOT CONTAIN ANTIFREEZE - FILLED WITH WATER AND INHIBITOR", shall be securely attached to radiator filler neck.

3.5.5.9.3 Antifreeze compound procedure. The cooling system shall be filled to the operating level with antifreeze compound conforming to MIL-A-11755. The compound shall be used without dilution. A warning tag bearing the information, "COOLING SYSTEM FILLED WITH ANTIFREEZE (ARCTIC-TYPE) - DO NOT DRAIN", shall be securely attached to radiator filler neck.

3.5.5.9.4 Cooling system security. After processing the cooling system as specified in 3.5.5.9, the armored radiator cap shall be secured with the combat lock.

3.5.5.10 Ramp winch assembly. All unpainted metal surfaces of the ramp winch assembly, excluding cylinder rod, shall be coated with preservative conforming to grade 4 of MIL-C-16173.

3.5.5.10.1 Ramp hydraulic reservoir. The ramp hydraulic reservoir shall be filled with hydraulic fluid conforming to MIL-H-6083.

3.5.5.11 Weapon/targeting station hydraulic system. The weapon/targeting station hydraulic system shall contain operational lubricant as specified on the applicable drawings, specifications, or lubrication order, filled to operating level.

3.5.5.12 Machine gun pintle mount. The machine gun pintle mount shall be removed and mounting bolts and washers reinstalled. All bare metal surfaces of the pintle shall be preserved with grease conforming to MIL-G-10924 and wrapped with a barrier material conforming to type II, grade A, class 2 of MIL-B-121. Wrap shall be secured with tape conforming to type III, class 1 of PPP-T-60. The preserved mount shall be identified and securely stowed within the vehicle.

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3.5.5.13 Hatches and doors. Rubber seals around hatches and doors shall be coated with powdered talc conforming to type IV, class C of ZZ-T-416, or technical talc conforming to MIL-T-50036. For shipment, the ramp and hatches shall be closed and secured from the inside. The engine access panel shall be secured with the combat lock located in the driver's compartment. The door located in the rear ramp will be used for the final exit. For level "A" shipments, the door shall be secured with a nut and bolt; with the nut welded to the bolt. For level "B" shipments, the nut shall not be welded. The thread of the bolt shall be deformed. For storage, hatches and doors shall be locked from the inside, except that the ramp door shall be secured in the open position for ventilation.

3.5.5.14 Engine compartment access components.

3.5.5.14.1 Engine compartment access plate and drain plugs. The engine compartment access plate and gasket with attaching hardware shall be removed for ventilation. A screen conforming to figure 2 shall be installed in the access plate opening and secured with four of the existing mounting screws and washers. The two forwarding drain plugs and the rear bilge drain plug shall be removed for drainage. A screen conforming to figure 3 shall be installed in the rear bilge opening and held in place with a spring retainer conforming to figure 4. Bare metal surfaces of the drain plugs shall be preserved with compound conforming to grade 4 of MIL-C-16173. Plate, gasket, preserved drain plugs, and four access plate mounting screws and washers shall be packaged as specified in 3.5.6.7. The information "REMOVE SCREENS, INSTALL ACCESS PLATE AND GASKET, FRONT AND REAR DRAIN PLUGS BEFORE VEHICLE OPERATION" shall be stenciled on the exterior of the vehicle using white or yellow paint conforming to MIL-P-13983. Stenciling shall be in characters not less than 3/4 inch [20 millimeters (mm)] high.

3.5.5.14.2 Engine compartment access panels. One engine compartment panel in the crew compartment shall be removed, and stowed securely in the crew compartment. A warning tag bearing the information "ENGINE PANEL REMOVED: LOWER RAMP OR OPEN HATCHES WHEN OPERATING ENGINE", shall be attached in a conspicuous location within the driver's compartment.

3.5.5.14.3 Level B: Engine compartment access panels. The engine compartment access panels in the crew compartment shall not be removed.

3.5.5.15 Miscellaneous preservation. Except as otherwise specified herein, all exposed, unpainted metal surfaces on the exterior of the vehicle, except the track shoes, shall be coated with compound conforming to grade 1 of MIL-C-16173. All exposed, unpainted, unplated metal surfaces on the interior of the vehicle shall be coated with compound conforming to grade 4 of MIL-C-16173.

3.5.6 Packaging.

3.5.6.1 Dry charge batteries and cables. Dry charged batteries and cables shall be installed and secured in the vehicle battery carrier. Battery cables shall be secured to the battery carrier with 3/4-inch (20 mm) tape conforming to type IV of PPP-T-97. Battery filler cap openings shall be sealed, by placing a 2-inch (50 mm) wide by 3 mil thick piece of film conforming to type II of MIL-B-22191 over each filler opening with the cap

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removed. The sheet shall be of sufficient length to allow it to be depressed into the opening to the same depth as the filler plug. Filler caps shall be screwed or inserted into openings, to form a complete seal without damaging the sheet. If batteries have been processed in accordance with MIL-B-11188, they need not be reprocessed as above.

3.5.6.2 Electrolyte. The electrolyte shall be packaged and packed in accordance with O-S-801, except that the exterior container shall conform to PPP-B-621, class 2, or PPP-B-601, overseas type. Marking shall be in accordance with O-S-801. The packed electrolyte shall be stowed in the same location as the BII and secured independently to permit separate removal.

3.5.6.3 Backrests and seats. Cushions of backrests and seats (see 3.5.4.1.2), shall be covered with paper conforming to type I, grade B of UU-P-268, with a minimum basic weight of 60 pounds [27 kilograms (kg)]. Paper shall be secured with tape conforming to type I of A-A-883.

3.5.6.3.1 Level B: Backrests and seats. Cushions of backrests and seats shall not be covered. If cushions are received packed, they shall be stowed as received in the crew compartment.

3.5.6.4 Squad leader's periscope and commander's viewing device. The squad leader's periscope and commander's viewing device shall be removed from the vehicle, cleaned, dried and immediately packaged and packed in accordance with Level A requirements of MIL-P-116 and MIL-P-14232, then securely stowed within the personnel compartment. The periscope opening shall be covered with a plate conforming to figure 5, item 3 and secured with four MS90725-59 screws.

3.5.6.4.1 Level B: Squad leader's periscope and commander's viewing device. The squad leader's periscope and commander's viewing device shall remain in place on the vehicle. The periscopes shall be cleaned in accordance with 3.5.4.2. The periscope lens shall be covered with hardboard and taped over in accordance with figures 6 and 7.

3.5.6.5 Fire extinguishers. Fire extinguishers shall contain a minimum of 90 percent of rated full charge. All seals shall be intact. DA Form 253 shall be completed, and attached securely to each extinguisher (see 6.3).

3.5.6.6 Basic issue items (BII). Unless otherwise specified (see 6.2), basic issue items shall be preserved, packaged, and packed in accordance with MIL-B-12841. BII shall be stowed and secured in accordance with 3.5.6.6.1.

3.5.6.6.1 Stowage and securement of BII. Basic issue items, and items removed for shipment, shall be identified and stored. BII shall be stored inside buildings (except during shipment). Packed BII, and removed items, shall be placed within the personnel compartment of the vehicle. Large wooden boxes shall be placed on the vehicle floor and shall be secured with 1-1/4 inch (32 mm) wide strapping conforming to type I, class 1, finish A of

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QQ-S-781. Strapping shall be secured to holding devices within the larger compartment. Smaller wooden boxes shall be secured to the larger boxes with 3/4-inch (20 mm) wide strapping conforming to type I, class 1, finish A of QQ-S-781. All boxes shall be secured in such manner as to prevent movement during transit, and to prevent damage to containers or the vehicle interior. Edge protectors shall be used under all strapping.

3.5.6.7 Access plate, gasket, and drain plugs. Access plate, gasket, and preserved drain plugs (see 3.5.5.14.1) shall be packed in a box conforming to type CF, class-weather-resistant of PPP-B-636. The box shall be closed with tape conforming to type III, class 1 of PPP-T-60, identified as to contents, and securely stowed within the personnel compartment.

3.5.6.8 Tow hooks. Tow hooks and related hardware shall be removed for shipment and packed in a box conforming to type CF, class-weather-resistant of PPP-B-636. The box shall be closed with tape conforming to type III, class 1 of PPP-T-60, identified as to contents, and securely stowed within the personnel compartment.

3.5.6.8.1 Level B: Tow hooks. Tow hooks shall be removed and stowed for overseas shipment only.

3.5.6.9 Radio antenna(s). Radio antennas shall be removed from the antenna mounts and wrapped in cushioning material conforming to type I of PPP-C-843, and placed in a fiberboard container. Store and secure the container within the vehicle.

3.5.6.10 PLRS antenna (M981 only). The PLRS antenna shall be removed from the vehicle and wrapped in cushioning material conforming to type I of PPP-C-843, and placed in a fiberboard container. Store and secure the container within the vehicle. The antenna opening shall be covered with a plate and gasket conforming to figure 5, items 1 and 2 and secured using four MS90725-59 screws.

3.5.6.11 Level B: Installed vision blocks. When vision blocks are left in place, proceed as follows: Blocks, shall be covered with hardboard conforming to figure 6. Each cover shall be cut to size and be covered with tape conforming to type IV, class 1, of PPP-T-60 or type II of MIL-T-22085.

3.5.6.12 Level B: Vision block openings. When vision blocks are not installed, plugs conforming to type I, class 2 of PPP-C-1752, 2 inches (50 mm) thick, 8-1/4 + 1/8 - 0 inches (210 + 3 - 0 mm) long and 2-1/4 + 1/8 - 0 inches (57 + 3 - 0 mm) wide, shall be installed in the vision block openings.

3.5.6.13 Level B: Bilge pump outlets. Bilge pump outlets shall be closed with plastic plugs conforming to MIL-C-5501/7. A warning tag, bearing the information, "BILGE PUMP OUTLETS CLOSED. PRIOR TO OPERATING THE BILGE PUMP, REMOVE PLUGS FROM OUTLETS", shall be secured to the bilge pump operating switch.

3.5.7 Height reduction for transportation. Unless otherwise specified (see 6.2), the launcher or targeting head shall be placed in the transport configuration position in accordance with the appendix of this specification.

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3.5.7.1 Level B: Height reduction for transportation. Unless otherwise specified (see 6.2), the launcher or targeting head shall be in the transport configuration for level "B" or driveaway requirements.

3.5.8 Vehicle closure.

3.5.8.1 Closure kit. When specified (see 6.2), each vehicle shall be provided with a vehicle protective closure kit. The closure kit shall be fabricated, assembled, and installed in accordance with Drawing 12269239.

3.5.8.1.1 Depot BII box marking. The information "DO NOT DESTROY - USE FOR SHIPMENT OF VEHICLE CLOSURE KIT" shall be stenciled on the depot BII box in lettering not less than 3/4-inch (19 mm) high, with a contrasting color of enamel conforming to TT-E-529. Do not mark the BII box if it is ascertained the container will not accommodate the closure kit.

3.5.8.2 Level B: Vehicle closure. Vehicle closure kits shall not be provided for level "B" shipment and storage. Openings to the vehicle interior shall be sealed with hardboard and tape in accordance with figures 6, and 7, except that overspray of the taping shall not be applied. The tape shall conform to type II of MIL-T-22085 or type IV, class 1 of PPP-T-60. The weapon and targeting stations shall be covered in accordance with figures 8 thru 11. Item 2 shall be secured in place wrapping girth-wise around the launcher or targeting head with PPP-S-760 strapping. Items 3 and 4 shall be secured in place by using two MS90725-22 hex screws (1/4-20 x 4), with four AN960-416 washers and two MS 51971-1 hex nuts. Item 1 shall be secured in place with two PPP-S-760 straps placed length-wise and one strap girth-wise around the launcher. Items 7 and 8 shall be secured in place by using two MS90725-18 hex screws (1/4-20 x 3), with four AN960-416 washers and two MS51971-1 hex nuts.

3.6 Loading.

3.6.1 Loading on flat cars. Loading of vehicles on open top railroad cars shall be in accordance with the applicable requirements of Section 1, General Rules Governing the Loading of Commodities on Open Top Cars, and figure 87 or 87A, Section 6, Rules Governing the Loading of Department of Defense Material on Open Top Cars, publication of the Association of American Railroads.

3.6.2 Highway shipment. Loading of vehicles for shipment by highway and rules for shipment by driveaway or towaway, shall be in accordance with Department of Transportation Publication, Federal Motor Carrier Safety Regulations, and applicable military publications.

3.6.3 Reprocessing engine after loading.

3.6.3.1 Level A. If the engine is operated in connection with the moving of the vehicle to a loading area or during the loading itself, the engine shall be reprocessed as specified in 3.5.5.5 through 3.5.5.5.5. The vehicle cover shall be rolled clear of the engine intake and exhaust to provide air circulation and to prevent damage to the cover. After reprocessing of the engine, the vehicle cover shall be replaced.

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3.6.3.2 Level B. If the engine is operated in connection with movement of the vehicle for loading or unloading, there shall be no additional processing of the engine.

3.7 Marking. Unless otherwise specified (see 6.2), the vehicle shall be marked in accordance with MIL-STD-129.

3.7.1 Lifting points. The legend "LIFT HERE", with an arrow pointing to the lifting eye, shall be stenciled adjacent to each lifting eye, using black ink conforming to TT-I-1795.

3.8 Drive-on - drive-off capability. When vehicles are to be operated for loading or unloading (see 6.2) the following provisions shall apply.

3.8.1 Additional fuel. When specified (see 6.2), a maximum of 10 gallons (38 liters) of fuel shall be added, to accomplish movement of the vehicle.

3.8.2 Batteries and electrolyte. Batteries shall be filled with electrolyte, be fully charged, and battery cables connected (see 3.5.6.1 and 3.5.6.2). After the vehicle self-movement for loading or placement in storage, the ground cable shall be disconnected at the battery and then secured to the battery carrier with 3/4 inch (20 mm) wide tape conforming to type IV of PPP-T-97. A tag bearing the information "VEHICLE PRESERVED FOR DRIVE-AWAY CONDITION. BEFORE CRANKING, CONNECT GROUND CABLE TO BATTERY TERMINAL. ENGINE AND FUEL TANKS NOT PRESERVED", shall be placed in a conspicuous location within the driver's compartment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection records. The contractor shall maintain records of all inspections performed and such records shall be readily available for review by the Government representative.

4.2 First article inspection. One of the first ten production processed vehicles (see 3.1), shall be subjected to the inspections specified in 4.5.

4.3 Inspection of production processed vehicles. Unless otherwise specified (see 6.2), one processed vehicle, selected at random from each days production shall be inspected in accordance with 4.5.2.1, 4.5.2.3, and 4.5.2.4.

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4.4 Rejection. Failure of the first article, or any production vehicle, to conform to the applicable requirements of this specification, shall be cause for rejection of the vehicle by the Government. The vehicle shall not be accepted until objective evidence that the contractor has corrected the condition causing the rejection has been provided to the Government.

4.5 Methods of inspection.

4.5.1 Materials. To determine conformance to 3.2, all materials to be used in processing of vehicles shall be inspected in accordance with the material specifications. Certified inspection and laboratory test reports shall be provided which show that furnished materials conform to the applicable material specifications, except for materials that have been inspected by the Government at source. Materials listed on a Qualified Products List shall be obtained from one of the approved sources indicated.

4.5.2 Processing. The vehicle processing shall be inspected to determine conformance to this specification. Inspection of processing shall include all requirements specified in table I and inspections 4.5.2.1 through 4.5.2.4.

4.5.2.1 Cleaning. The interiors of the vehicles shall be examined for cleanliness to determine conformance to 3.5.4.1. One vehicle each day shall be tested for cleanliness in accordance with the applicable provisions of MIL-P-116. To determine conformance to 3.5.4.2, the exterior of the vehicle shall be examined for cleanliness. The surfaces on which tape is to be applied, shall be examined for cleanliness before application.

4.5.2.2 Engine. To determine conformance to 3.5.5.5.1 through 3.5.5.5.5, the interior of the engine from one of the first ten production processed vehicles shall be examined for surface coverage. The engine shall be disassembled to the extent necessary to permit visual examination of all surfaces within the combustion chamber. The combustion chamber includes all the surfaces within the cylinder from, and including the surface of the head within the combustion chamber. All surfaces within the combustion chamber shall have a wet coating of preservative oil, as obtained when the item is dipped or flushed with the oil. The processing method of the approved preserved engine shall be submitted in writing for use in processing engines in all subsequent production vehicles.

4.5.2.3 Fuel tank. To determine conformance to 3.5.5.8, a visual inspection of preservative application shall be accomplished to assure addition of the correct amount of oil, based upon a known amount of residual fuel.

4.5.2.4 Cooling system. To determine conformance to 3.5.5.9, one processed vehicle shall be selected at random from each day's production for inspection. The engine coolant shall be tested using a hydrometer-thermometer type tester with a range of minus 60°F (-50°C) to plus 106°F (+40°C) conforming to GG-T-250.

5. PACKAGING

This section is not applicable to this specification.

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TABLE I. Processing inspection.
(see indicated paragraphs for level A, and B, requirements)

	Cleaning	Preservation		Packaging/Stowage	
	Levels A and B	Level A	Level B	Level A	Level B
Preparation prior to processing				3.5.1	3.5.1
Processing records				3.5.1.1	3.5.1.1
Disassembly				3.5.2	3.5.2
Record forms				3.5.3	3.5.3
Cleaning and drying	3.5.4				
Vehicle interior	3.5.4.1				
Battery supports and retainers	3.5.4.1.1				
Backrests and seats	3.5.4.1.2			3.5.6.3	3.5.6.3.1
Vehicle exterior	3.5.4.2				
Preservation		3.5.5	3.5.5		
Relubrication		3.5.5.1	3.5.5.1		
Preservation of battery supports and retainers		3.5.5.2	3.5.5.2		
Transmission, transfer assembly, controlled differential and final drives*		3.5.5.3	3.5.5.3.1		
Engine crankcase preservation*		3.5.5.4	3.5.5.4.1		
Engine preservation*		3.5.5.5	3.5.5.5.6		
Preservation through fuel system and combustion chamber		3.5.5.5.1	3.5.5.5.6		
Engine purging		3.5.5.5.2	3.5.5.5.6		
Preservation through air intake and exhaust system*		3.5.5.5.3	3.5.5.5.6	3.5.5.5.3	
Preservation through oil level gage rod opening		3.5.5.5.4	3.5.5.5.6	3.5.5.5.4	
Preservation through flywheel housing		3.5.5.5.5			
Air cleaner				3.5.5.6	
Personnel and engine compartment heaters and lines		3.5.5.7	3.5.5.7.1	3.5.5.7	
Fuel tank preservation		3.5.5.8	3.5.5.8.2		
Cooling system*		3.5.5.9	3.5.5.9		
Water and anti-freeze procedure		3.5.5.9.1	3.5.5.9.1		

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TABLE I. Processing inspection - Continued.
(see indicated paragraphs for level A, and B, requirements)

	Cleaning	Preservation		Packaging/Stowage	
	Levels A and B	Level A	Level B	Level A	Level B
Water and corrosion inhibitor procedure		3.5.5.9.2	3.5.5.9.2		
Antifreeze compound procedure		3.5.5.9.3	3.5.5.9.3		
Ramp winch assembly		3.5.5.10	3.5.5.10		
Ramp hydraulic reservoir		3.5.5.10.1	3.5.5.10.1		
Weapon/targeting station hydraulic system		3.5.5.11	3.5.5.11		
Machine gun pintle mount		3.5.5.12	3.5.5.12	3.5.5.12	3.5.5.12
Hatches and doors		3.5.5.13	3.5.5.13		
Ventilation		3.5.5.14	3.5.5.14		
Engine compartment access plate and drain plugs		3.5.5.14.1	3.5.5.14.1	3.5.6.7	3.5.6.7
Engine compartment access panels				3.5.5.14.2	3.5.5.14.3
Miscellaneous preservation		3.5.5.15	3.5.5.15		
Dry charge batteries and cables				3.5.6.1	3.5.6.1
Electrolyte				3.5.6.2	3.5.6.2
Periscopes	3.5.4.2			3.5.6.4	3.5.6.4.1
Fire extinguishers				3.5.6.5	3.5.6.5
Basic issue items				3.5.6.6	3.5.6.6
Tow hooks				3.5.6.8	3.5.6.8.1
Radio antenna(s)				3.5.6.9	3.5.6.9
PLRS antenna				3.5.6.10	3.5.6.10
Height reduction for transportation				3.5.7	3.5.7.1
Vehicle closure kit				3.5.8.1	3.5.8.2
Depot BII box marking				3.5.8.1.1	
Loading on flat cars				3.6.1	3.6.1
Highway shipment				3.6.2	3.6.2
Reprocessing engine after loading		3.6.3.1	3.6.3.2		
Marking				3.7	3.7
Lifting points				3.7.1	3.7.1

* Inspect DD Form 1397

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6. NOTES

6.1 Intended use. Vehicle processing covered by this specification is intended to protect the vehicles for storage outside of buildings, immediate use, domestic or overseas shipment, and including carloading.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable level of processing (see 1.2).
c. If first article inspection is not required (see 3.1).
d. Applicable procedure for protection of cooling system (see 3.5.5.9).
e. If BII shall be processed, packed or stored other than as specified (see 3.5.6.6).
f. If launcher or targeting head transport configuration shall be other than as specified (see 3.5.7).
g. If closure kit is required (see 3.5.8.1).
h. If draining of residual fuel from fuel tank is required for Level B processing (see 3.5.5.8.2).
i. If vehicle markings shall be other than specified (see 3.7).
j. If vehicle drive-on/drive-off capability is required (see 3.8).
k. If additional fuel shall be supplied (see 3.8.1).
l. If responsibility for inspection shall be other than as specified (see 4.1).
m. If inspection shall be other than as specified (see 4.3).

6.3 Safety precautions. Caution should be exercised in handling carbon dioxide (CO_2) fire extinguisher cylinders. Cylinders should not be dropped, permitted to strike each other, or be handled roughly. Extreme care should be exercised during the reinstallation operation to avoid tripping the fire extinguisher control trigger (see 3.5.6.5).

6.4 Forms. A copy of the "Equipment Log Book" and all required forms (see 3.5.3) will be furnished to the contractor by the Government at least 30 days before shipment of the vehicles required by the contract delivery schedule.

6.5 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 4.5.1).

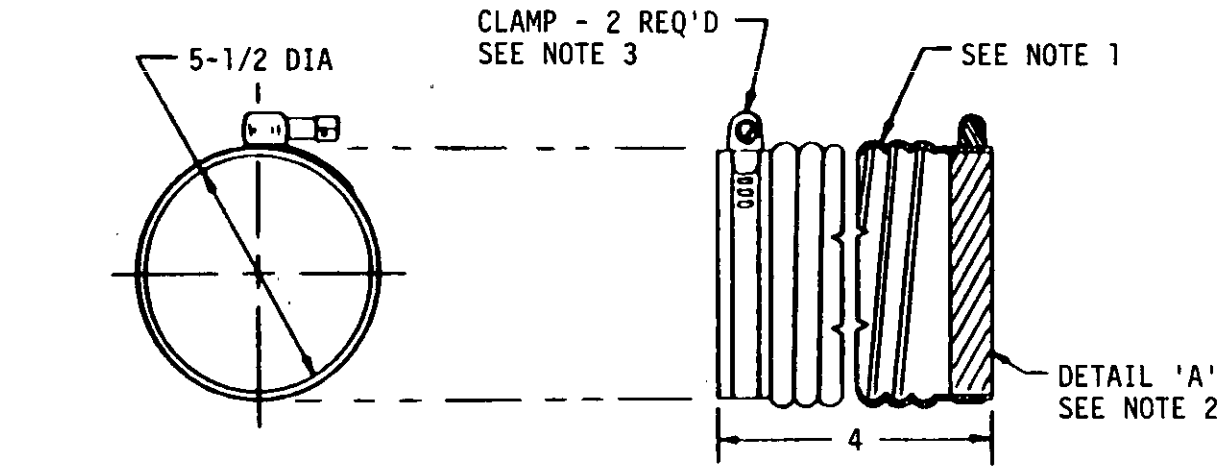
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

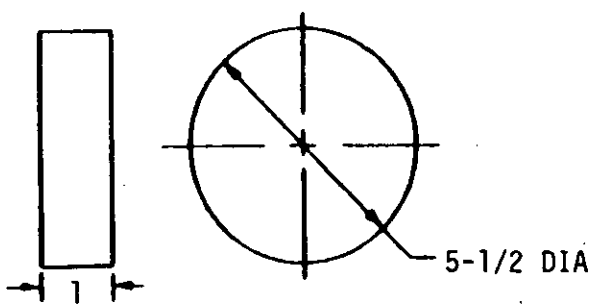
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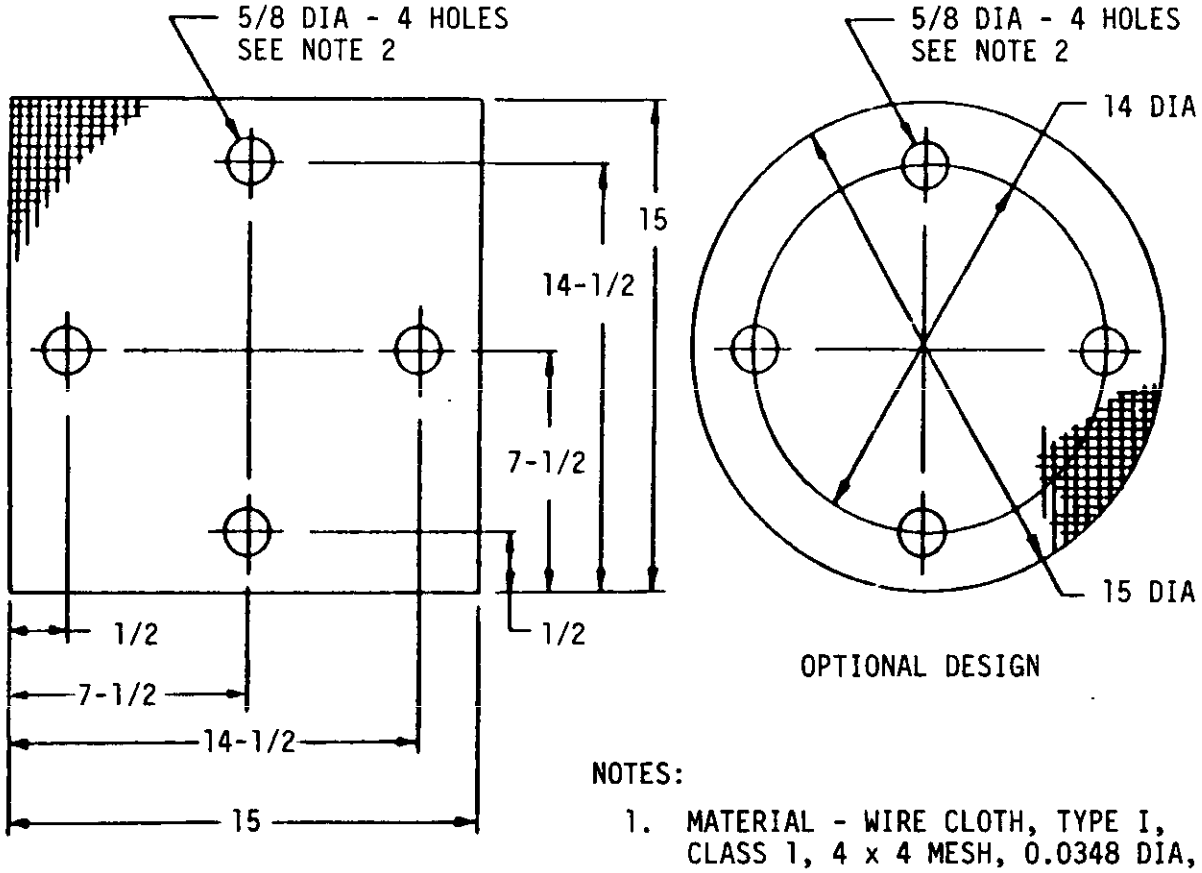
DETAIL 'A' - 1 REQ'D

NOTES:

1. FLEXIBLE HOSE TO BE MADE FROM MATERIAL CONFORMING TO MIL-H-52079 OR EQUIVALENT.
2. PLUG TO BE MADE FROM 1 INCH PLYWOOD, NN-P-530, STD INT WITH EXT GLUE.
3. USE MS35842-16 OR EQUIVALENT CLAMPS.
4. DIMENSIONS ARE IN INCHES.

FIGURE 1. Boot, air restrictor.

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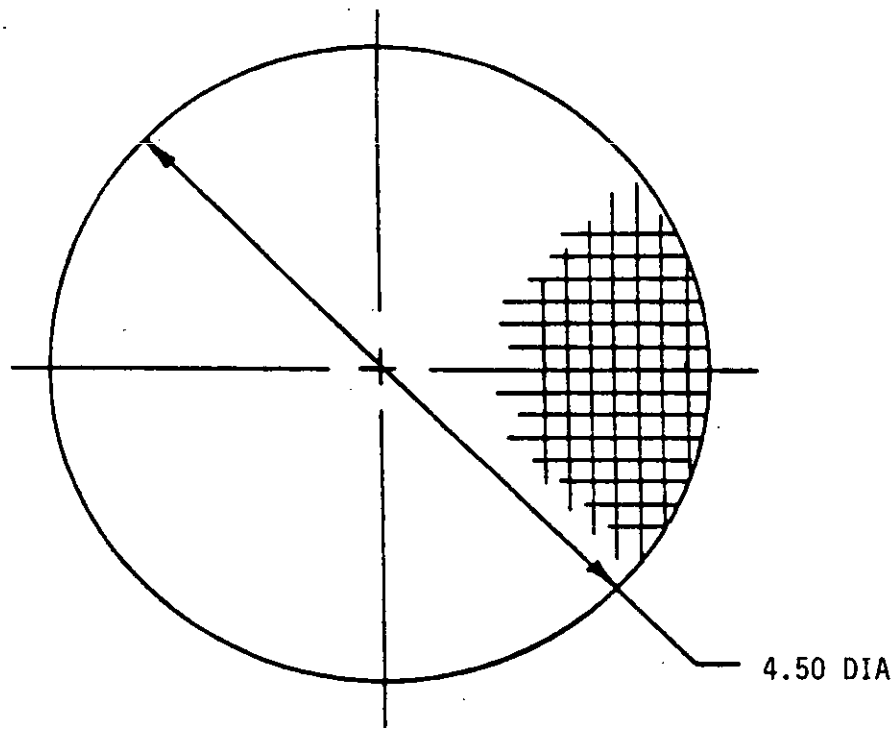


NOTES:

1. MATERIAL - WIRE CLOTH, TYPE I, CLASS 1, 4 x 4 MESH, 0.0348 DIA, PER SPEC RR-W-360.
2. HOLES MAY BE FORMED WITH DRIFT PIN AT INSTALLATION.
3. ALL DIMENSIONS ARE IN INCHES.

FIGURE 2. Screen.

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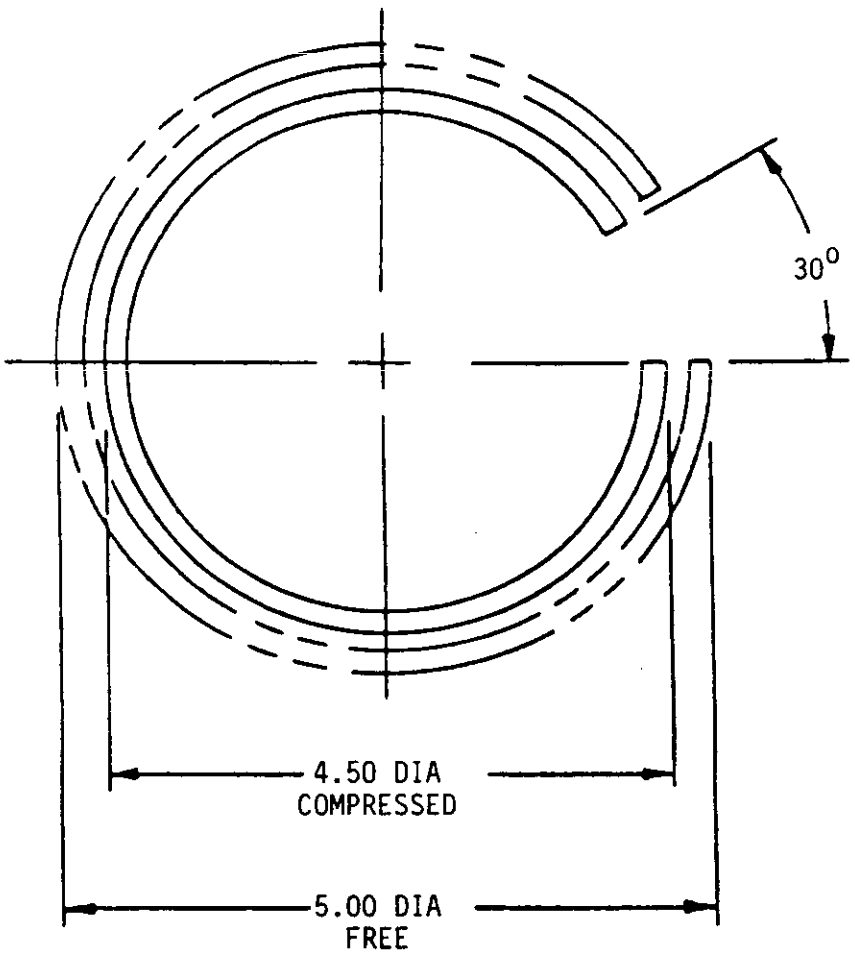


NOTES:

1. MATERIAL - WIRE CLOTH, TYPE I, CLASS 1, 4 x 4 MESH, 0.0348 DIA, PER SPEC RR-W-360.
2. DIMENSIONS ARE IN INCHES.

FIGURE 3. Screen.

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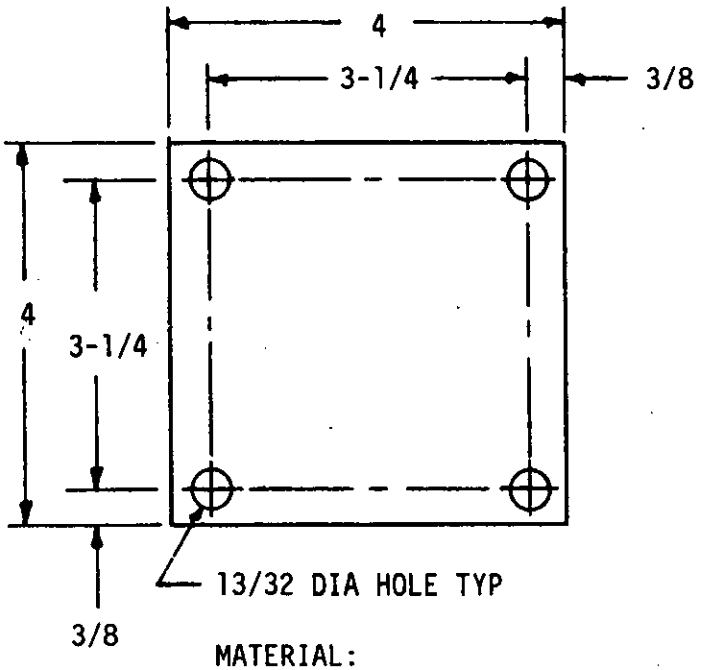


NOTES:

1. MATERIAL - WIRE, STEEL, CARBON, SPRING, MUSIC, PER SPEC QQ-W-470, 0.156 DIA.
2. DIMENSIONS ARE IN INCHES.

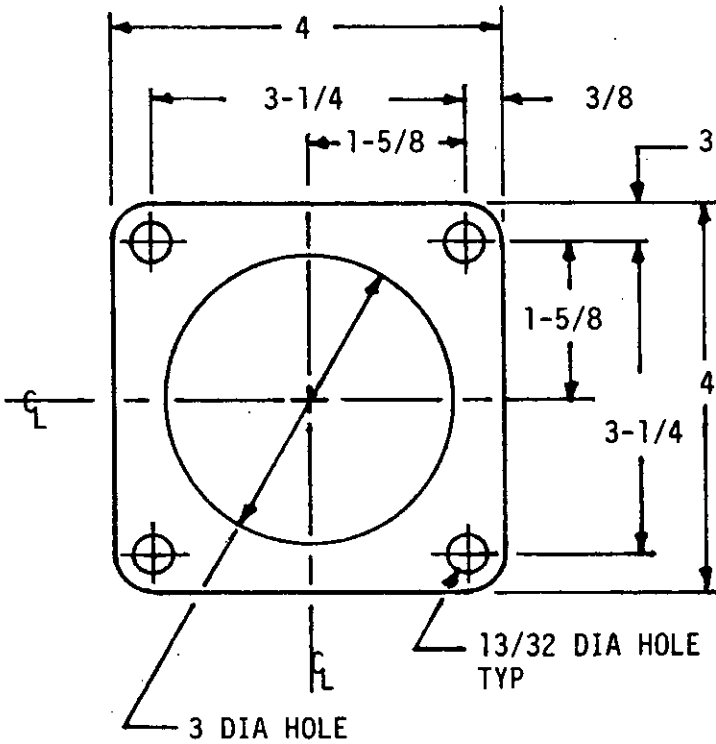
FIGURE 4. Spring, retainer.

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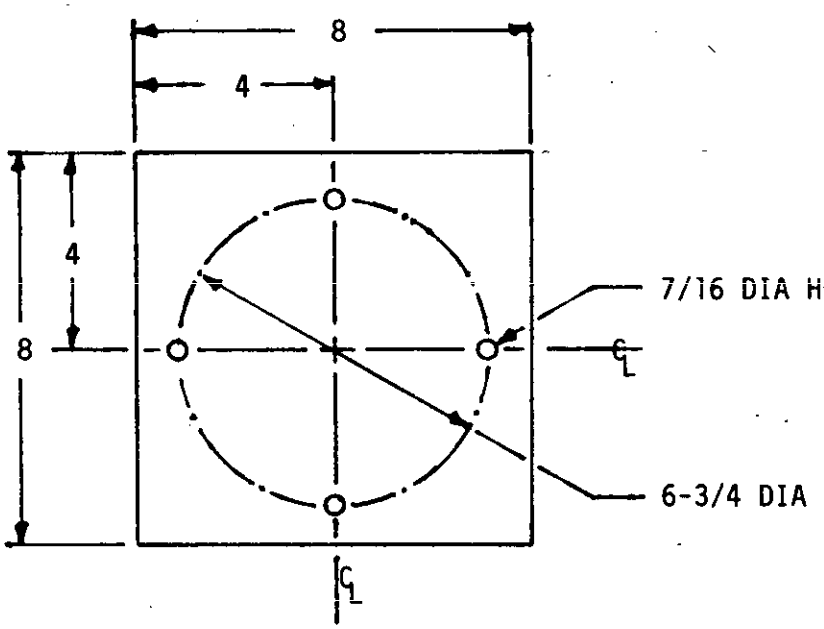
MATERIAL:
ALUMINUM ALLOY 6061-0
PER SPEC QQ-A-250/11
0.063 THICK

ITEM 1



MATERIAL:
RUBBER, GRADE RN/RN510 B.C.F.,
PER SPEC MIL-R-3065
0.063 THICK

ITEM 2



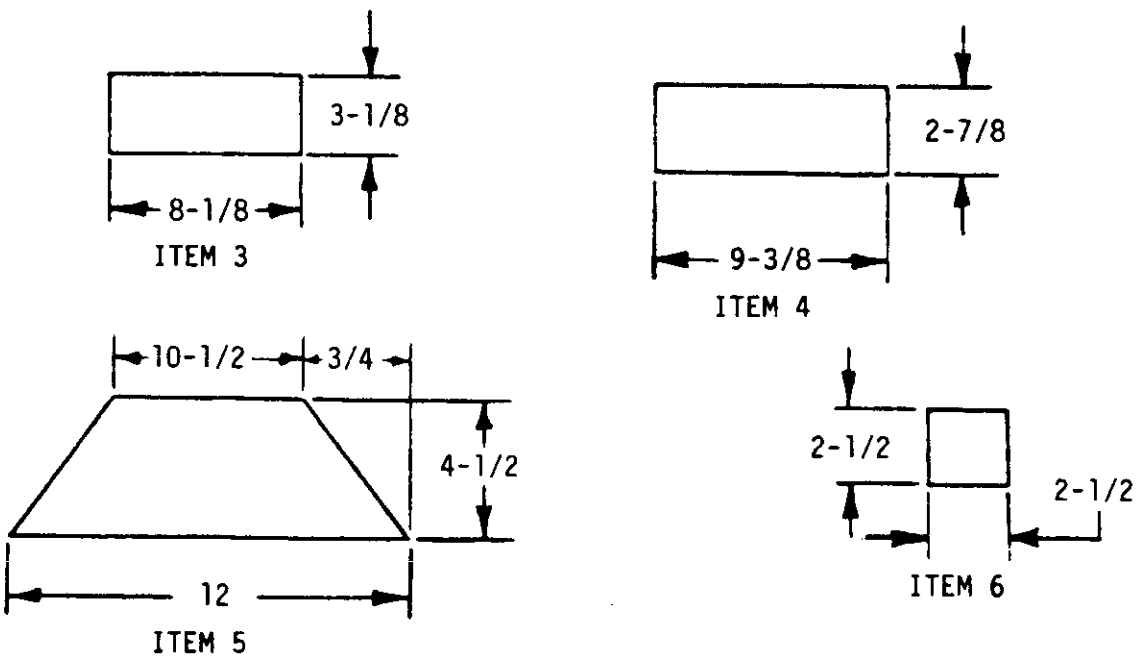
ITEM 3

NOTE: ALL DIMENSIONS ARE IN INCHES.

MATERIAL:
ALUMINUM ALLOY 6061-0
PER SPEC QQ-A-250/11
0.063 THICK

FIGURE 5. Closures.

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MATERIAL:

HARDBOARD, LLL-B-810, TYPE II, SURFACE II, FINISH D, DESIGN A, 1/8 INCH THICK.

SMOOTH ROUGH EDGES.

ALL DIMENSIONS ARE IN INCHES, TOLERANCE $\pm$ 1/8.

FIGURE 6. Hardboard panels.

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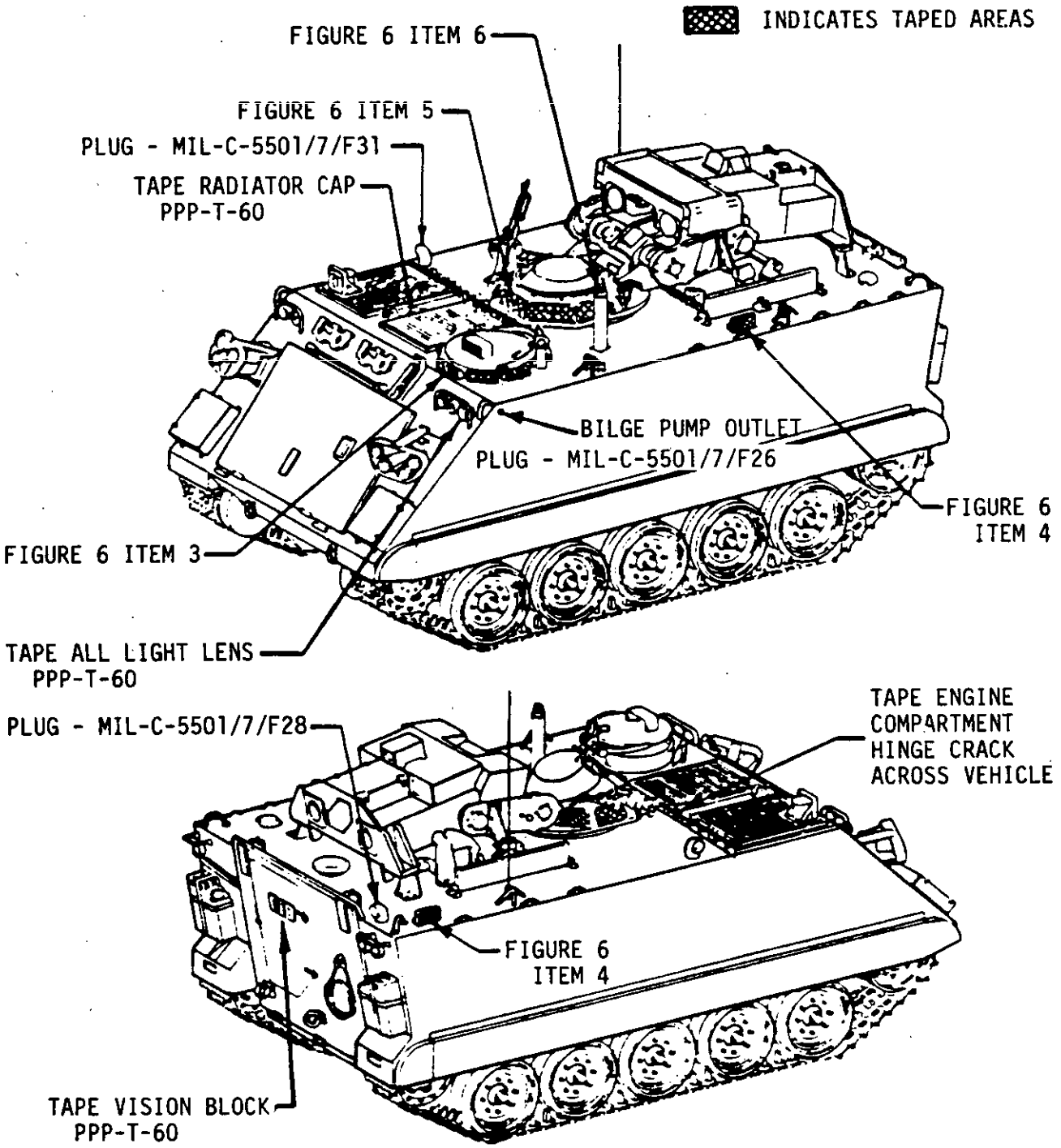
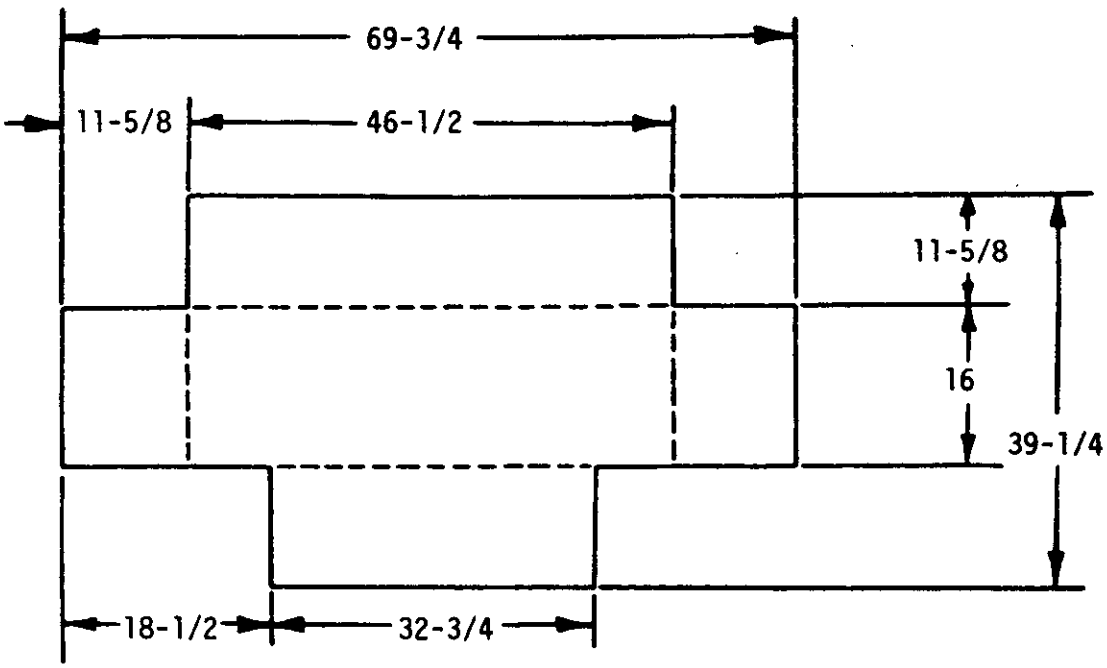


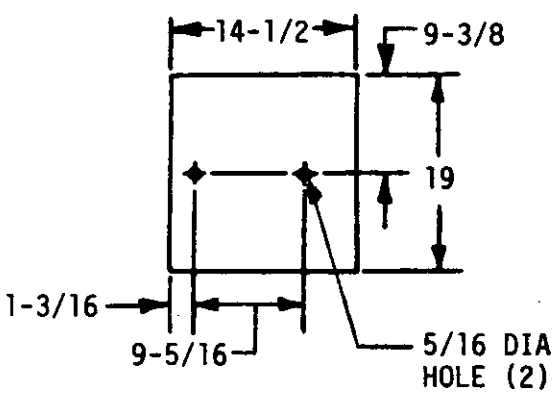
FIGURE 7. Level 'B' closures.

(M901/M901A1 vehicle shown above.
M981 vehicle closures are the same.)

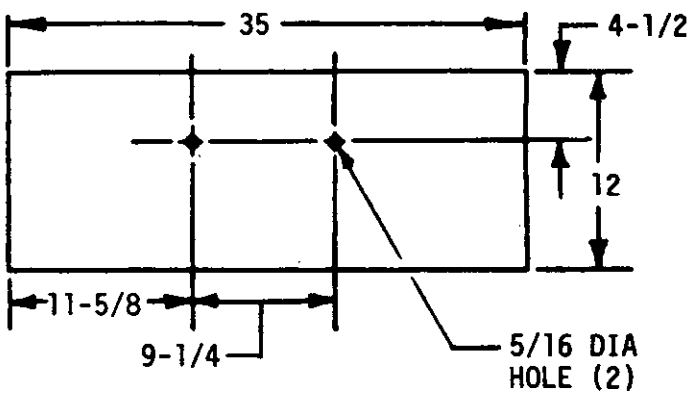
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ITEM 1



ITEM 3
2 EA



ITEM 4

NOTES:

1. ITEMS 1, 3, AND 4 - TYPE CF, CLASS 2, WEATHER-RESISTANT, VARIETY TW (TRIPLE-WALL), PER SPECS PPP-F-320 AND PPP-B-640.
2. ITEM 2 (NOT SHOWN) - BOX 11-1/4 x 9-1/2 x 5-7/8, TYPE CF, GRADE V3C, STYLE: HALF-SLOTTED, PER SPEC PPP-B-636.
3. ALL DIMENSIONS ARE IN INCHES.

FIGURE 8. M901/M901A1 - level 'B' closures.

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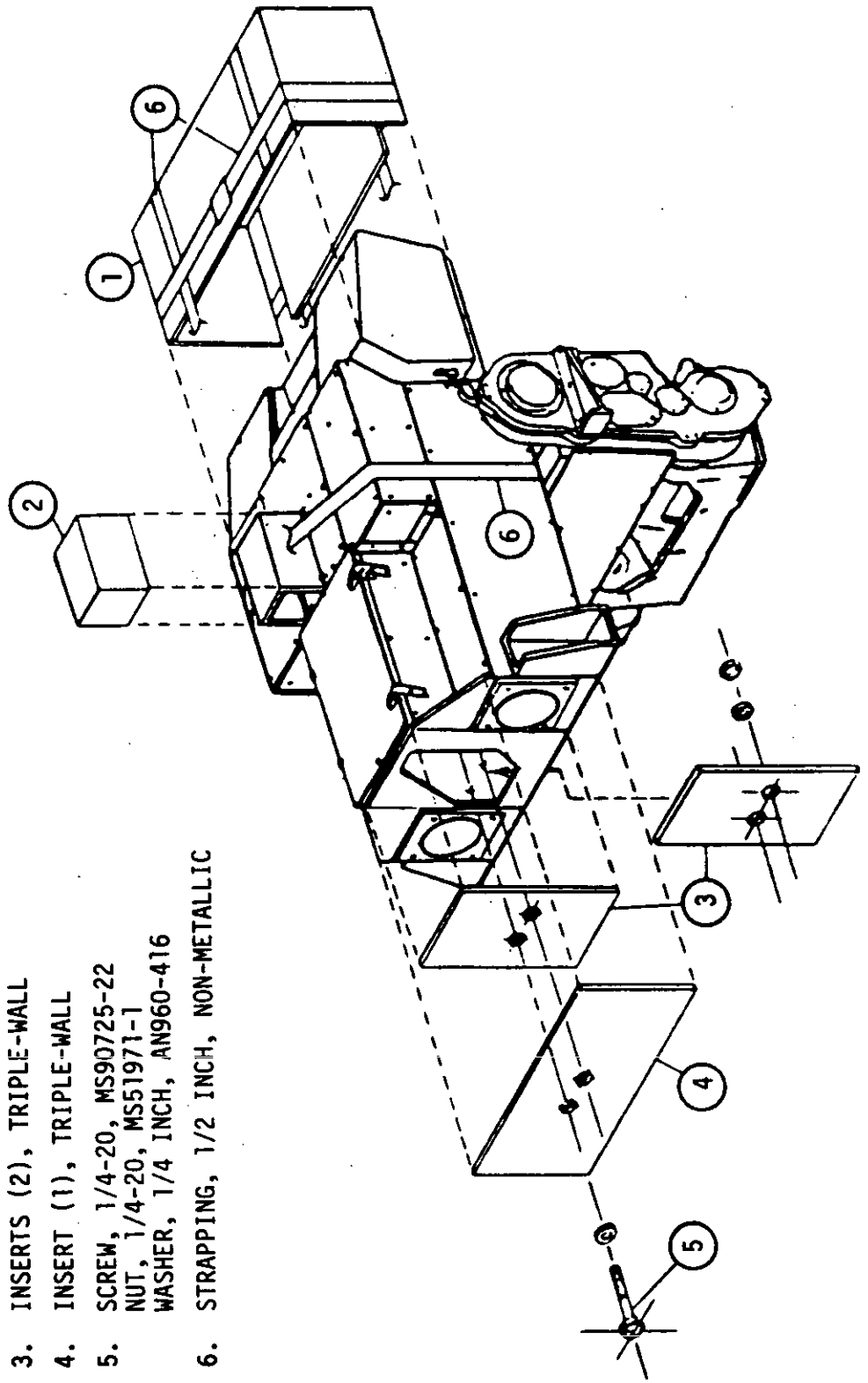
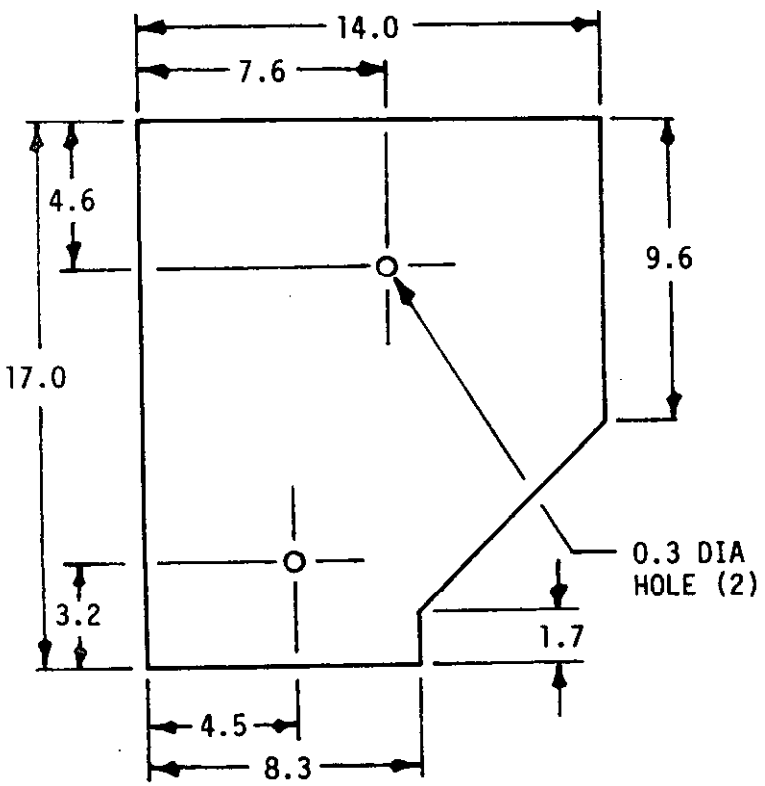
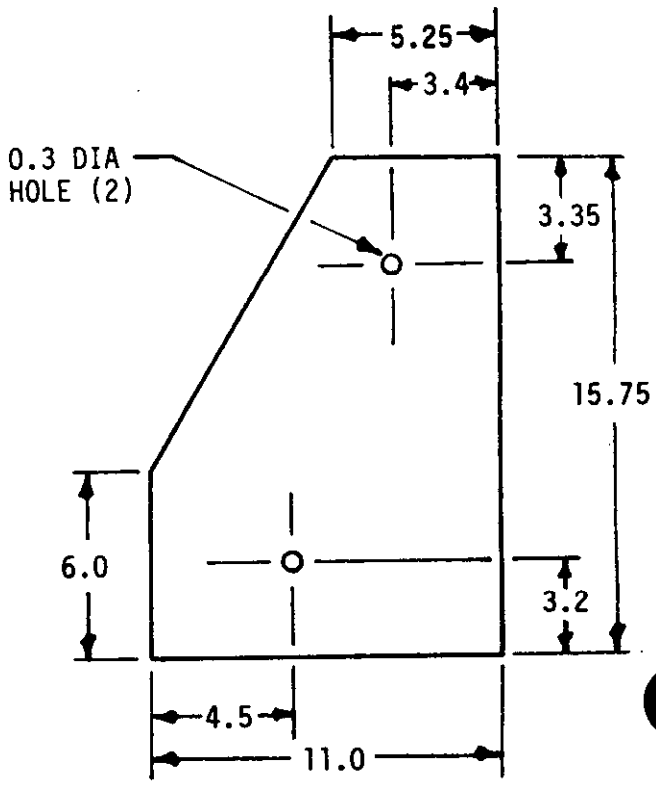


FIGURE 9. Level 'B' closure.

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ITEM 7



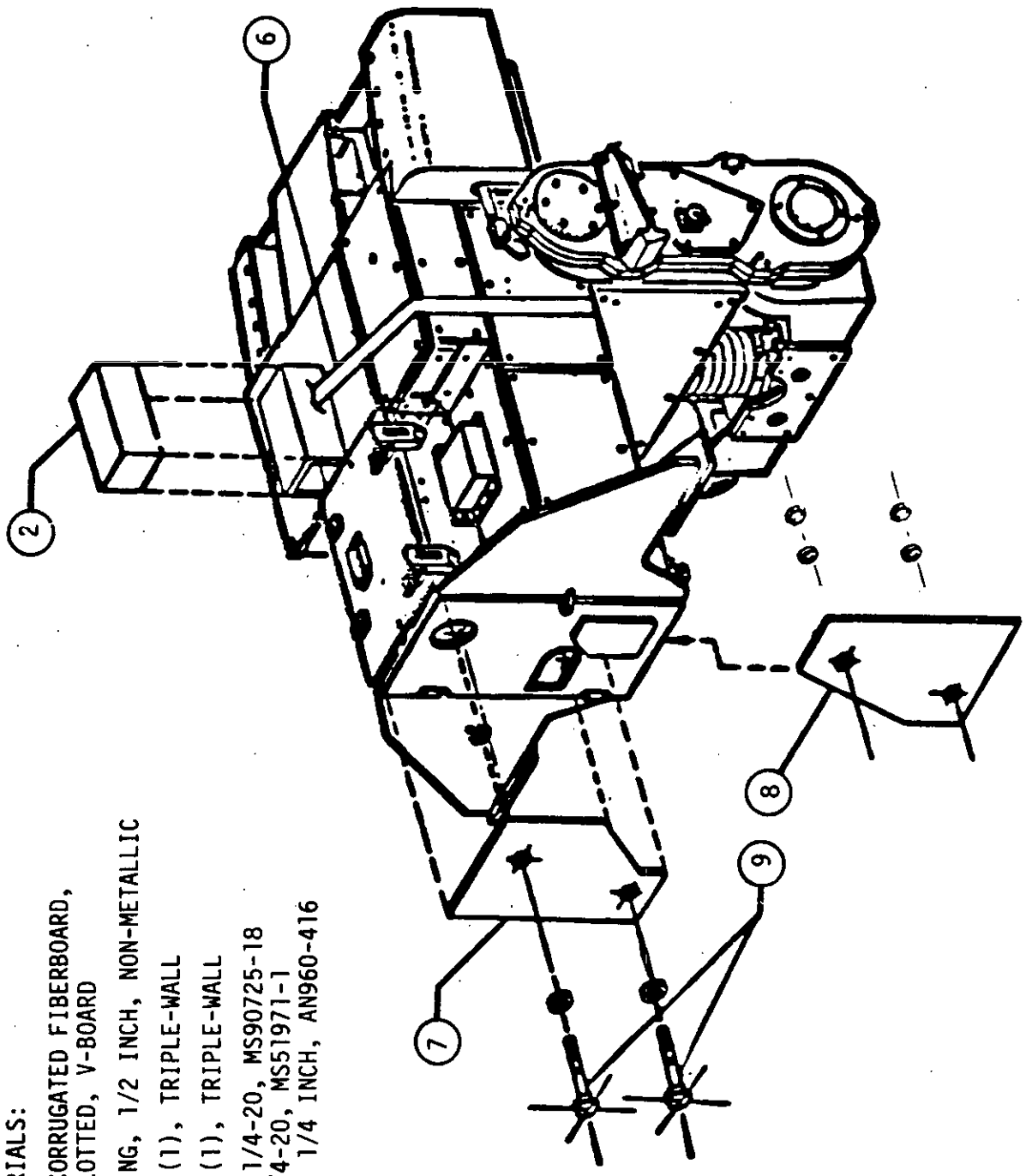
ITEM 8

NOTES:

1. ITEMS 7 AND 8 - TYPE CF, CLASS 2, WEATHER-RESISTANT, VARIETY TW (TRIPLE-WALL), PER SPECS PPP-F-320 AND PPP-B-640.
2. ITEM 2 (NOT SHOWN) - BOX 11-1/4 x 9-1/2 x 5-7/8, TYPE CF, GRADE V3C, STYLE: HALF-SLOTTED, PER SPEC PPP-B-636.
3. ALL DIMENSIONS ARE IN INCHES.

FIGURE 10. M981 - level 'B' closure.

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Check the source to verify that this is the current version before use.



BILL OF MATERIALS:

2. BOX - CORRUGATED FIBERBOARD, HALF-SLOTTED, V-BOARD
6. STRAPPING, 1/2 INCH, NON-METALLIC
7. INSERT (1), TRIPLE-WALL
8. INSERT (1), TRIPLE-WALL
9. SCREW, 1/4-20, MS90725-18
   NUT, 1/4-20, MS51971-1
   WASHER, 1/4 INCH, AN960-416

FIGURE 11. M981 - Level 'B' closure.

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APPENDIX

REQUIREMENTS FOR CONVERTING THE LAUNCHER AND THE TARGETING HEAD FROM STANDARD TO TRANSPORT CONFIGURATION, AND BACK TO STANDARD

10. SCOPE

10.1 Scope. This appendix covers the requirements for reducing the M901/M901A1 and M981 vehicle overall height for transport.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. REQUIREMENTS

30.1 Standard to transport configuration conversion procedure. To reduce the M901/M901A1 and M981 vehicle overall height for transport, proceed as follows: Place launcher/targeting head in erect position. Bleed system down as specified in 30.2 or 30.3 as applicable. Remove stowage brackets as specified in 30.4.2 and proceed through 30.5.

30.2 Hydraulic pressure bleed-down procedure (M901/M901A1 only). To bleed off hydraulic pressure, proceed as follows:

a. Apply power to turret in the normal manner to place launcher in erected positions at 180 degrees in azimuth (facing rear).
b. Set EMER PWR switch to UP position.
c. Move hand controller up and down in elevation, until launcher ceases motion.
d. Set TURRET POWER switch to OFF, and then set EMER PWR switch to OFF.
e. Ensure that ERECTION LOCKS are engaged. If not, repeat steps a through d, to ensure erection locks remain engaged.
f. Support front of launcher, using two C-clamps (NSN 5120-00-203-6431) (2) and two 2 x 4 lumber bracing (1) as shown in figure 12.

30.3 Hydraulic pressure bleed-down procedure (M981 only). To bleed off hydraulic pressure, proceed as follows:

a. Apply power to turret in the normal manner to place the targeting head in erect position at least 180 degrees in azimuth.
b. Set PUMP switch on TSCD to DISABLE.
c. Move hand controls up and down in elevation in small increments until targeting head ceases motion.
d. Set TSCD PWR switch to OFF and set PUMP switch to ON.
e. Ensure that lock cylinders are engaged.
f. Support front of targeting head, using two C-clamps (NSN 5120-00203-6431) (2) and two 2 x 4 lumber bracing (1) as shown in figure 12.

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30.4 Removal of items for transport configuration.

30.4.1 Legend for figures 13, 14 and 15.

1. Bolt (4)
2. Washer (4)
3. Lockwasher (4)
4. Launcher/targeting head stow support
5. Bolt (2)
6. Washer (2)
7. Stow Position Stop
8. Bolt (2)
9. Washer (2)
10. Deleted
11. Nuts (2)
12. Bumper
13. Shim
14. Transport Pad
15. Bolt (6)
16. Lockwasher (6)
16.1 Washer (6)
17. High Stow Bracket
18. Bolt (7)
19. Washer (7)
20. WFOV Telescope
21. ITA Cover
22. Bolt (2)
23. Washer (2)
24. Elevation Indicator
25. Washer (4)
26. Elevation Scale
27. Bolt (3)
28. Ventilator Cover
29. Spacer (3)

30.4.2 Bracket removal procedure. The procedure listed hereafter shall be followed to remove stowage brackets:

a. Remove 4 bolts (1), 4 washers (2), 4 lockwashers (3) and launcher/targeting head stow support (4). Repeat for the other side.
b. From the rear of erection arm, remove bolts (5), 2 washers (6), and stow position stop (7). Repeat for the other side.
c. Remove 2 bolts (8), 2 washers (9), 2 nuts (11), bumper (12) and shim (13). Repeat for the other side.
d. Apply corrosion preventive compound (MIL-C-16173, grade I) to 2 bolts (8).
e. Install to transport pad (14) on loading hatch: shim (13), bumper (12), 2 washers (9), and 2 bolts (8). Repeat for the other side.
f. Lower the side armor (M901/M901A1 only) and remove 6 bolts (15), 6 lockwashers (16), 6 washers (16.1) and erection arm bumper bracket (17). Repeat for the other side.
g. Identify all the removed hardware, pack in a box conforming to type CF, class WR of PPP-B-636. Box closure with tape conforming to type III, class 1 of PPP-T-60. Stow within the vehicle.

30.4.3 Removal of ITA cover and WFOV telescope. To remove the cover and telescope, proceed as follows (see figure 9b):

a. Remove 7 bolts (18), 7 washers (19), and ITA cover (21). Replace bolts and washers and pack cover in a box conforming to type CF, class WR of PPP-B-636. Stow within the vehicle.
b. Loosen split ring retainer from the top of the large ring, and then unscrew the large ring on the base of WFOV telescope. Remove the telescope (20). Cover the telescope opening with plastic conforming to type II, class 1 of L-P-378. Secure with tape conforming to type IV, class 1 of PPP-T-60 or type

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II of MIL-T-22085. Wrap the telescope with paper conforming to NNN-P-40. Wrap lens with material conforming to PPP-C-843. Secure with tape conforming to type III, class 1 of PPP-T-60. Place the telescope in a box conforming to type CF, class WR of PPP-B-636. Box closure with tape conforming to type III, class 1 of PPP-T-60. Identify box as to contents and stow within the vehicle.

30.4.4 Removal of elevation indicator.

a. Remove 2 bolts (22), 2 washers (23), elevation indicator (24), and 4 washers (25) from upper inboard side of right-hand erection arm.
b. Position 2 washers (23) on 2 bolts (22) and reinstall 2 bolts (22) to the full depth on the erection arm.
c. Identify all removed hardware, pack in a box conforming to type CF, class WR of PPP-B-636. Box closure with tape conforming to type III, class 1 of PPP-T-60. Stow within the vehicle.

30.4.5 Removal of ventilator cover (M981 only).

a. Remove 3 bolts (27), ventilator cover (28) and 3 spacers (29).
b. Cover the ventilator opening by installing the cover (1), bracket (2) and spacer (3) with the MS90725-124 bolt (4) (see figure 12).
c. Identify all removed hardware, pack in a box conforming to type CF, class WR of PPP-B-636. Box closure with tape conforming to type III, class 1 of PPP-T-60. Stow within the vehicle.

30.4.6 Removal of forward plate (M981 only). To remove the forward plate assembly, refer to figure 16, and proceed as follows:

Legend for figure 16:

1. Cotter Pin (2)
2. Flat Washer (2)
3. Straight Headed Pin (2)
4. Erection Plate Assembly
5. Machine Screw (6)
6. Erection Arm Guide (2)
7. Erection Arm Shim (2)

a. Remove 2 cotter pins (1), 2 flat washers (2), 2 straight headed pins (3), and forward plate assembly (4).
b. Remove 6 machine screws (5), 2 erection arm guides (6) and 2 erection arm shims (7).
c. Identify all the removed hardware, pack in a box conforming to type CF, class WR of PPP-B-636. Box closure with tape conforming to type III, class 1 of PPP-T-60. Stow within the vehicle.

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30.5 Launcher/targeting head crank down procedure. To lower launcher/targeting head proceed as follows:

a. Apply system power, remove 2 x 4 lumber bracing and C-clamps, and then place launcher/targeting head to the elevation stowed position as indicated by the elevation stow light.
b. For M901/M901A1 only, set MODE SELECT switch to STOW.
c. For M981 only, set HEAD switch to STOW.
d. Use the erection drive switch to move the launcher/targeting head to the high stow position.
e. Remove cap (1) from the erection arm (see figure 17).

CAUTION: All personnel should stand clear of the handcrank. Handcrank may occasionally jump abruptly.

f. Install handcrank (2) and turn handcrank clockwise to crank launcher/targeting head down to the deck.
g. Remove handcrank, install cap, and turn off the system power.
h. Reduce the height of the radio antenna(s) in accordance with 3.5.6.9 and 3.5.6.10.

30.6 Transport to standard configuration conversion procedure. To place the M901/M901A1 or M981 vehicle into standard configuration, from the transport configuration, refer to figures 13, 14 and 15, and proceed as follows:

a. M901/M901A1 only. Set the TURRET POWER switch to ON, and MODE switch to ERECT. Use the ERECTION DRIVE switch to erect launcher. Bleed down hydraulic system pressure, and support the launcher as specified in 30.2.
b. M981 only. Set the TSCD PWR switch to ON and HEAD switch to ERECT. Use the DRIVE switch to erect targeting head. Bleed down the hydraulic system pressure and support the targeting head as specified in 30.3.
c. Install WFOV telescope (20), ITV cover (21), 7 washers (19), and 7 bolts (18) (see figure 15). Adjust WFOV telescope (20) as follows: Without disturbing the ITA diopter ring, select the ITA wide field-of-view channel (3X position on the sight select switch) for a clear sharp image of the wide field-of-view circular reticle.

   Lock-wire the wide field-of-view focusing ring, and the optical sight and night sight channel objective lens focusing rings in place.

CAUTION: Be sure to put split ring retainers back into position.

d. Apply corrosion preventive compound (grade 1, MIL-C-16173) to 6 bolts (15).
e. Install stow bracket (17), 6 washers (16.1), 6 lockwashers (16), and 6 bolts (15). Repeat for the other side (see figure 14). Torque bolts (15) to 50-55 foot-pounds.

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f. Remove 2 bolts (8), 2 washers (9), bumper (12), and shim (13) from the low stow cushion base (14) on cargo hatch (see figure 14). Repeat for the other bumper.
g. Apply corrosion preventive compound (grade 1, MIL-C-16173) to 2 bolts (5).
h. Install shim (13), bumper (12), 2 washers (9), 2 bolts (8), and 2 nuts (11) to the high stow bracket (17). Repeat for the other bumper.
i. Apply corrosion preventive compound (grade 1, MIL-C-16173) to 2 bolts (5).
j. Install stow position stop (7), 2 washers (6), and 2 bolts (5) to rear of the erection arm. Repeat for the other stow position stop. Torque bolts (5) to 8 to 9 foot-pounds.
k. Apply corrosion preventive compound (grade 1, MIL-C-16173) to 4 bolts (1).
l. Install the launcher/targeting head support (4), 4 washers (3), 4 lockwashers (2), and 4 bolts (1). Repeat for the other launcher/targeting head support. Torque bolts (1) to 25 to 30 foot-pounds.
m. Remove bolt (4), cover (1), bracket (2) and spacer (3) (see figure 18).
n. Install 3 spacers (29), ventilator cover (28), and 3 bolts (27) (see figure 15).
o. Remove 2 bolts (22) and 2 washers (23).
p. Install 4 washers (25), elevation indicator (24), 2 washers (23), and 2 bolts (22).

NOTE

Do not tighten bolts (22).

q. Adjust the elevation indicator (24) as follows:

NOTE

Clinometer must face the same direction for measurements taken in steps (1) and (2) below.

1. Place a clinometer on a flat surface of the launcher/targeting head and note the angle indicated and the color of the scale.
2. Place a clinometer on a flat surface of the vehicle top deck, and note the angle indicated and the color of the scale.
3. If the color of the scales are the same (both black or both red), subtract the angle of the deck obtained in step (2) from the angle of the launcher/targeting head obtained in step (1). The difference is the elevation angle that the elevation pointer (24) and scale (26) should show. If the color of the scales is different (one red and one black), add the angle obtained in step (2) to the angle obtained in step (1). The sum is the elevation angle that the elevation pointer (24) and elevation scale (26) should show.

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4. Position the elevation pointer (24) so that correct launcher/targeting head elevation angle is shown on the elevation scale (26) and tighten 2 bolts (22).

r. Install 2 erection arm shims (7), 2 erection arm guides (6), and 6 machine screws (5) (see figure 16).
s. Install erection plate assembly (4) with 2 straight headed pins (3), 2 flat washers (2), and 2 cotter pins (1).

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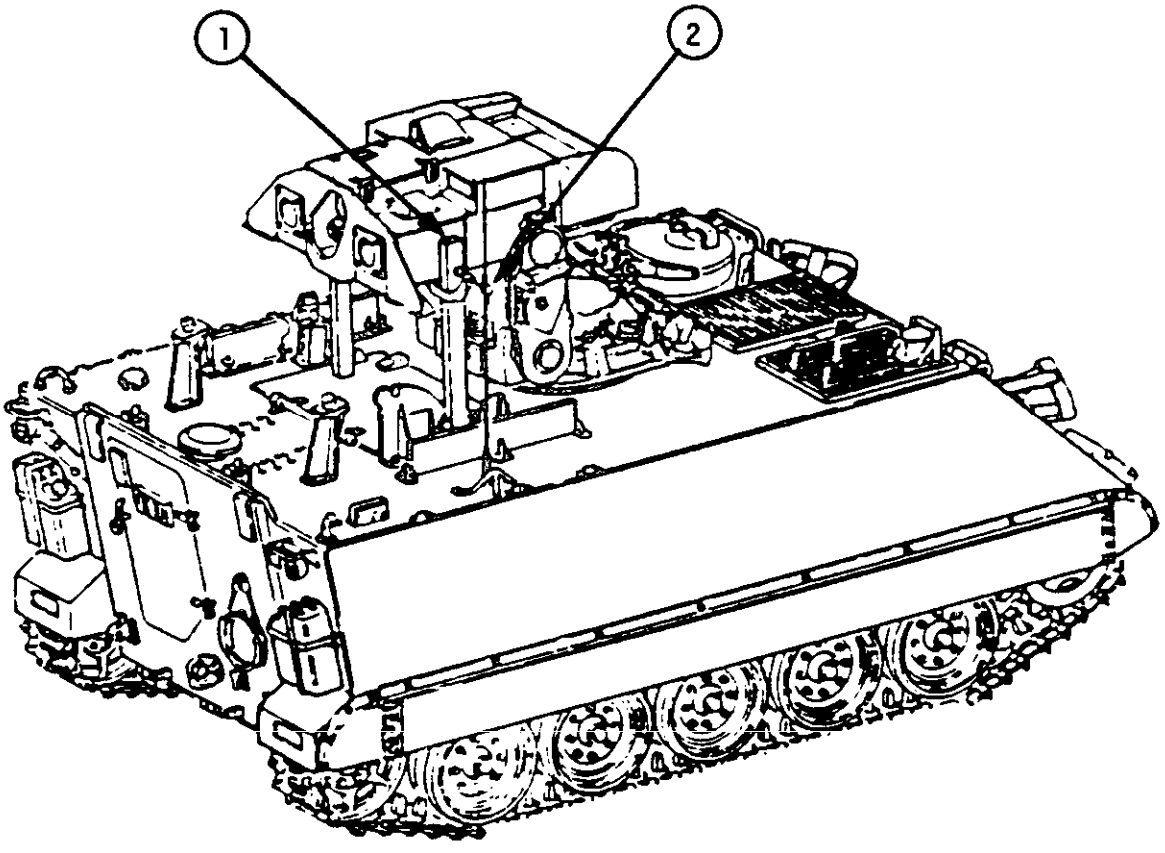


FIGURE 12. Hydraulic pressure bleed down.

(This figure is applicable for the M901/M901A1 and M981.)

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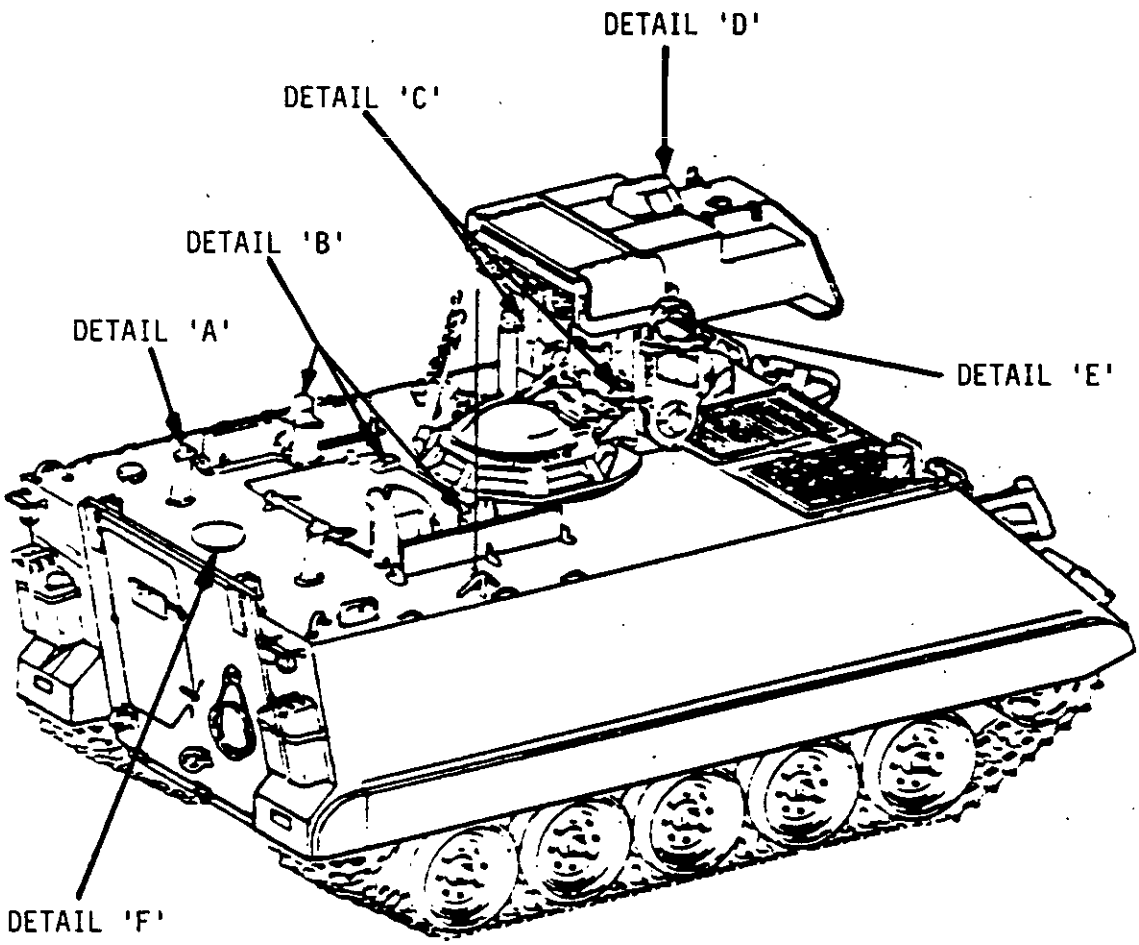


FIGURE 13. Vehicle conversion (transport).

(This figure is applicable for the M901/M901A1 and M981.)

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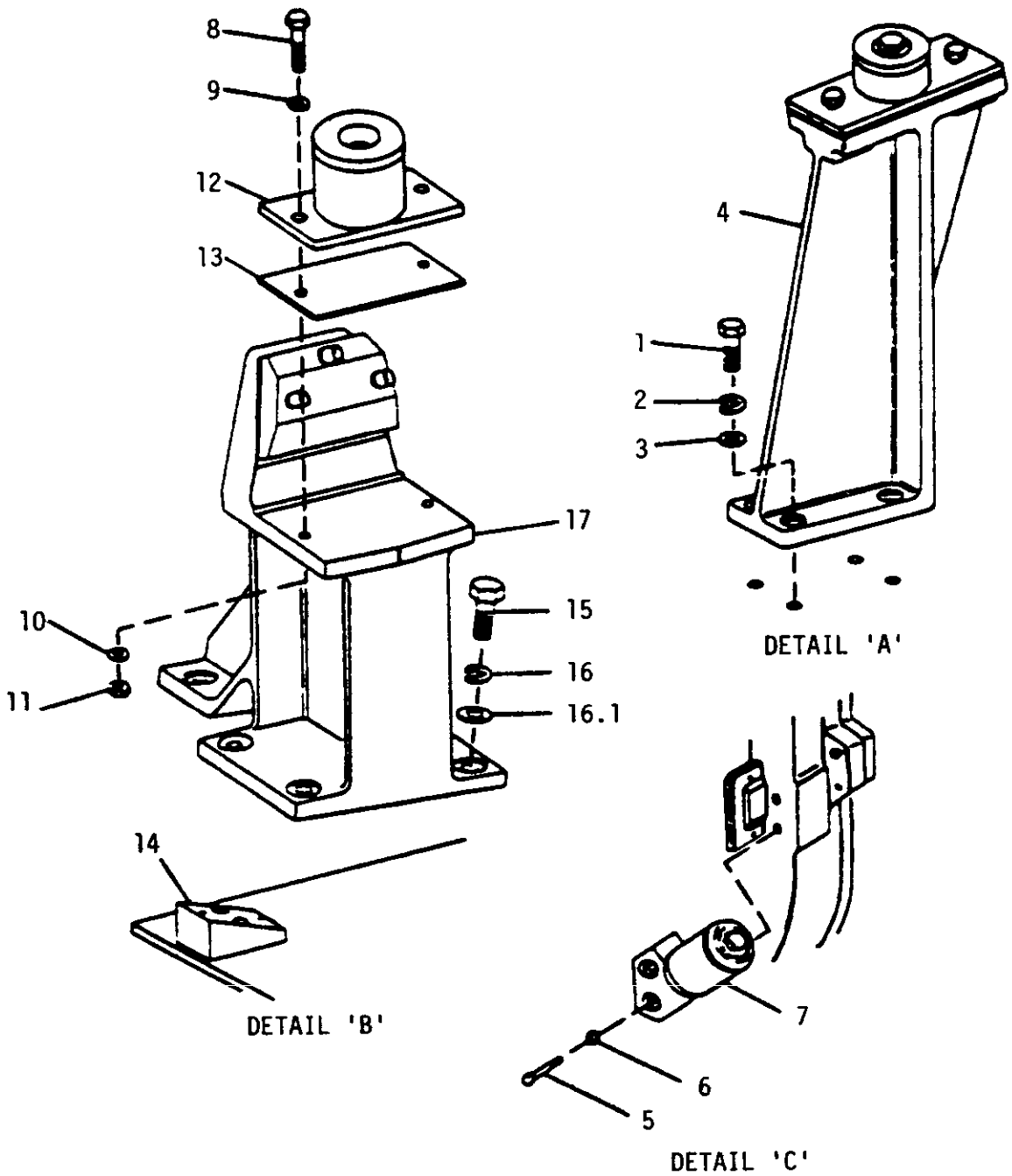


FIGURE 14. Vehicle conversion (transport).

(This figure is applicable for the M901/M901A1 and M981.)

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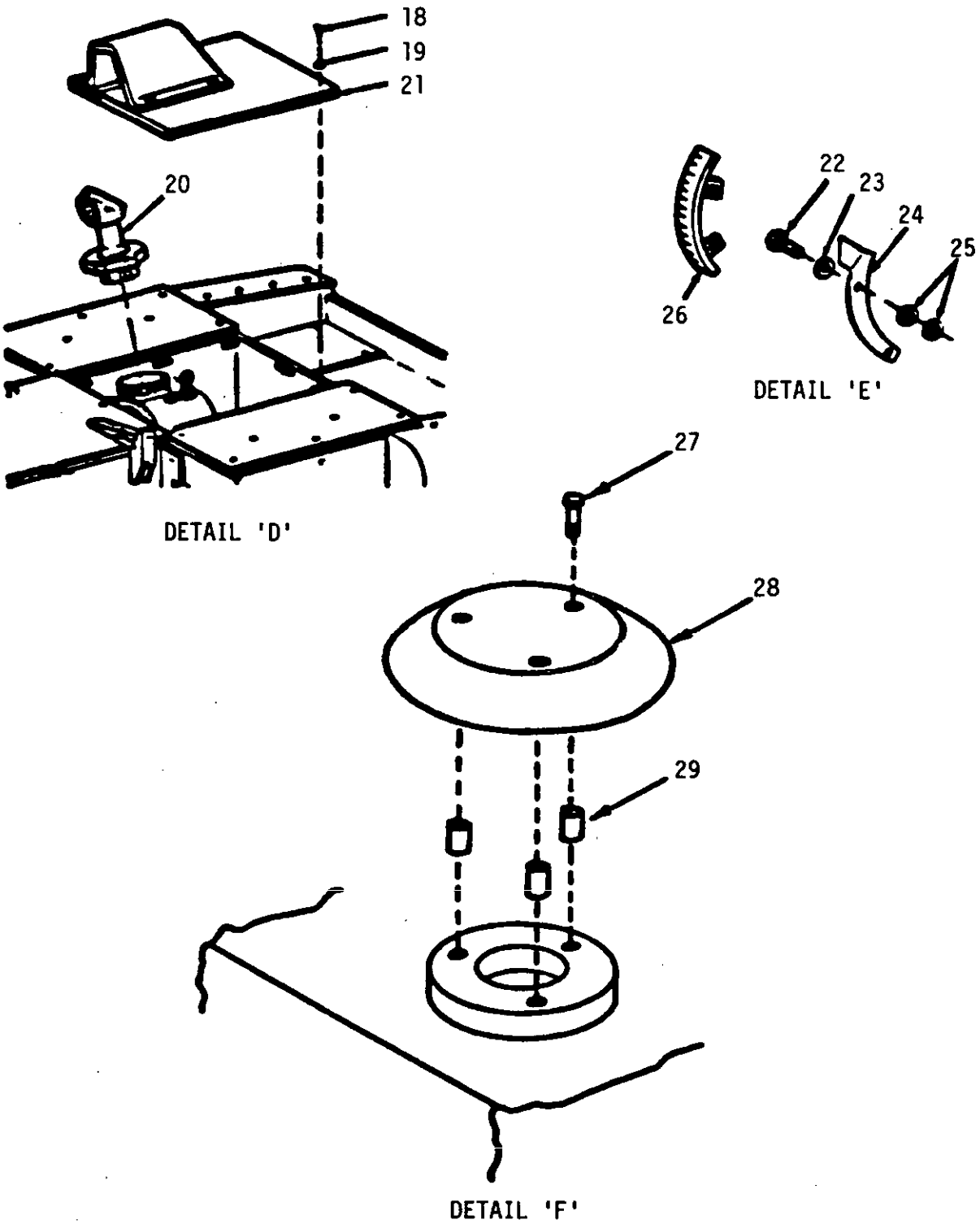


FIGURE 15. Vehicle conversion (transport).

(This figure is applicable for the M901/M901A1 and M981.)

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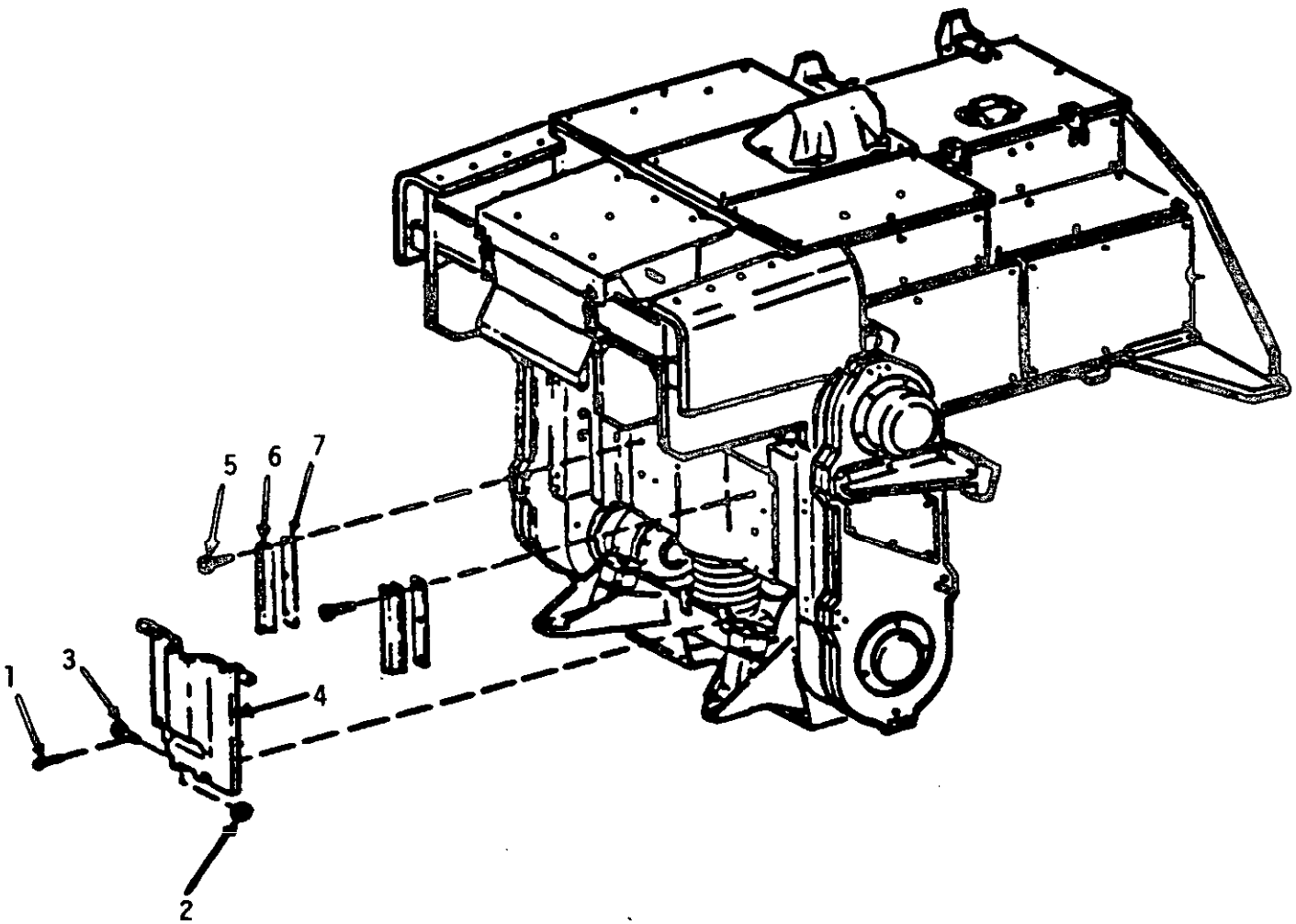


FIGURE 16. Forward plate removal and installation.
(This figure is applicable for the M981 only.)

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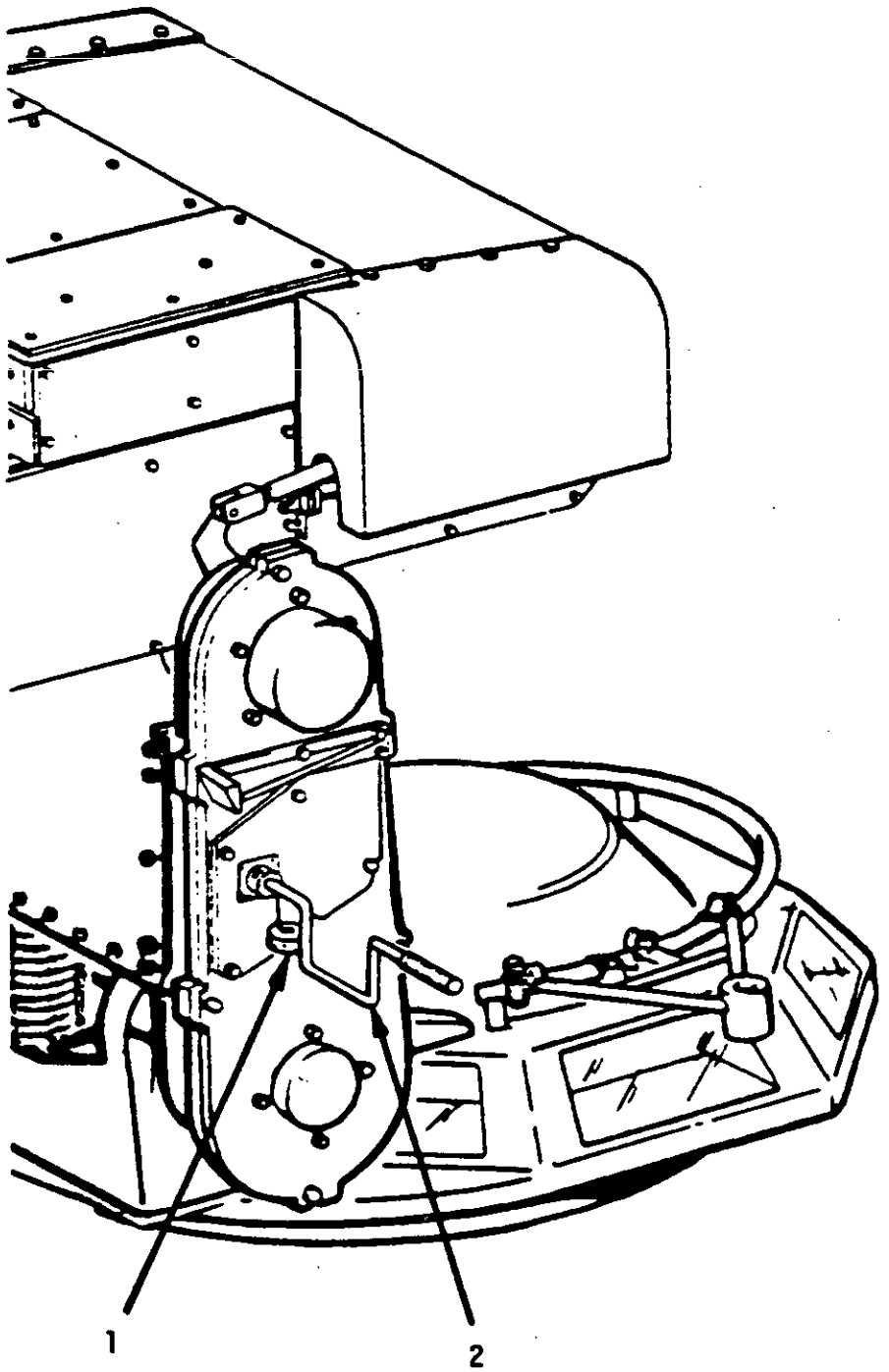
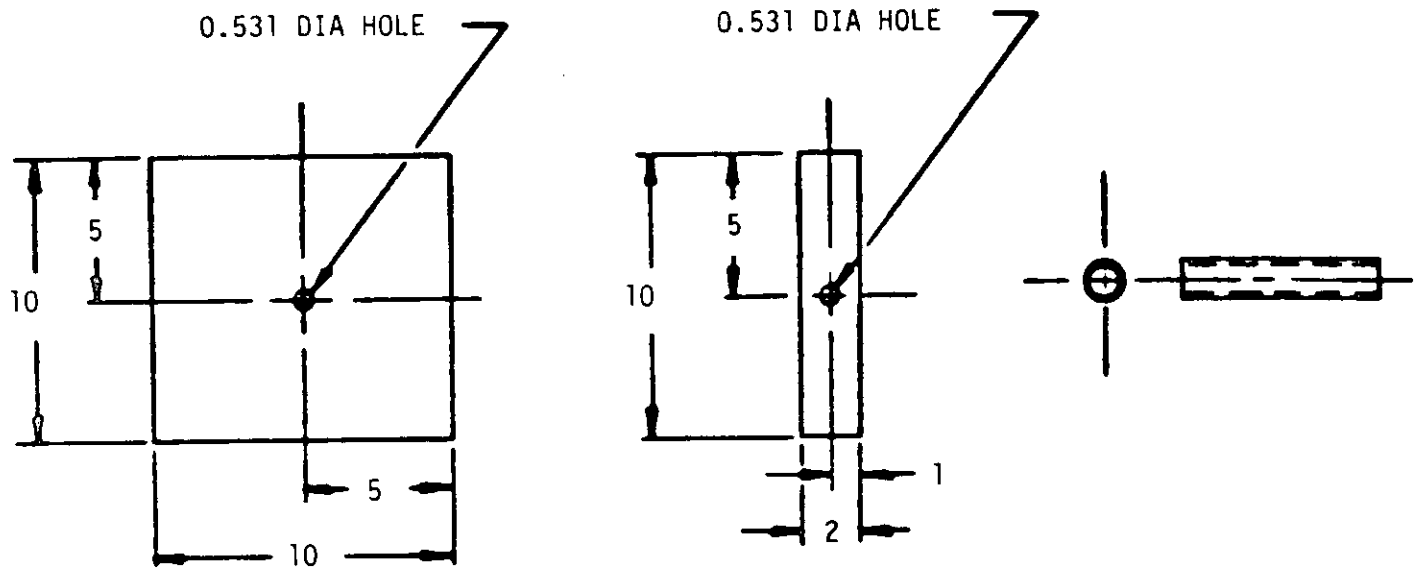


FIGURE 17. Installation of handcrank.

(This figure is applicable for the M901/M901A1 and M981.)

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COVER

MATERIAL:
ALUMINUM ALLOY 2024-T4
PER SPEC QQ-A-250/4
0.125 THICK

ITEM 1

BRACKET

MATERIAL:
ALUMINUM ALLOY 2024-T4
PER SPEC QQ-A-250/4
0.125 THICK

ITEM 2

SPACER

NAS43DD8-188

ITEM 3

NOTE:
ALL DIMENSIONS ARE IN INCHES.

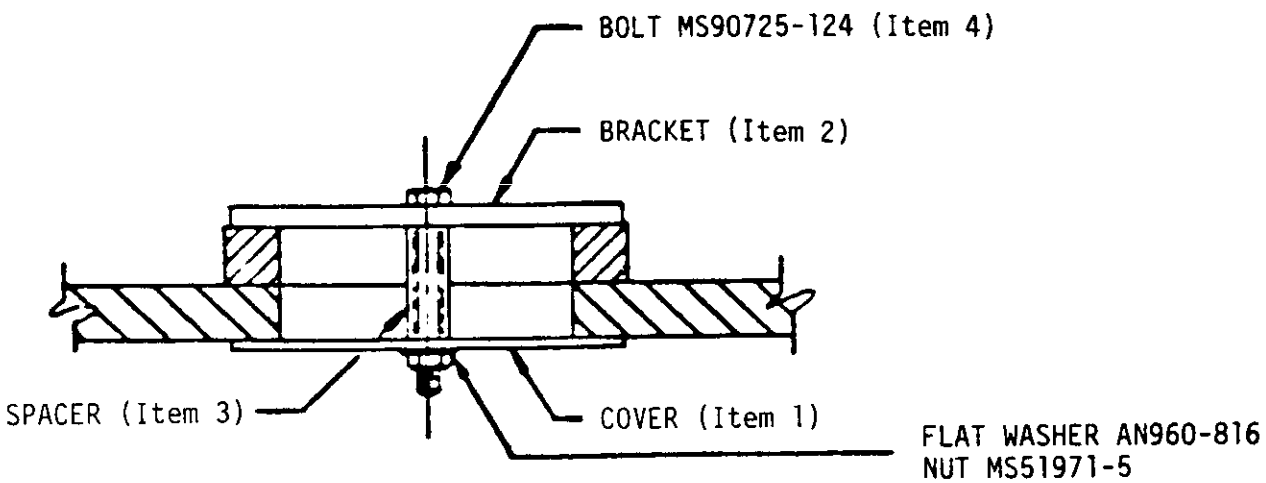


FIGURE 18. Ventilator hole closure.

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